PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 4

Cost and Management Accounting

Revision Cards

Every flashcard of every lesson, chapter by chapter: the question, then its answer. Cover the answer, say it, then check.

711 cards from 66 lessons · phone edition · made 2026-10-20 from the PassBook lessons

The labs (step-through simulators) are not printed: open the lesson in the PassBook app for the interactive version.

Contents

1 Introduction to Cost and Management Accounting 7

1.1 Objectives, scope and users of cost and management accounting 7

1.2 Cost accounting department and installing a costing system 10

1.3 Cost, financial and management accounting, and financial management 13

1.4 Cost units, cost centres, cost objects and responsibility centres 15

1.5 Elements and classification of cost 19

1.6 Cost behaviour: fixed, variable, semi-variable and step costs 21

1.7 Cost control and cost reduction 24

1.8 Methods and techniques of costing, and digital costing 27

2 Cost Sheet 30

2.1 Cost sheet for a manufacturer 30

2.2 Estimated cost sheets, tenders and quotations 32

2.3 Cost sheet for the service sector 35

3 Material Cost 38

3.1 Procurement procedure, GeM, ERP and stock verification 38

3.2 Valuing material receipts 41

3.3 Pricing material issues: FIFO, LIFO and average methods 43

3.4 Stock levels: re-order, minimum, maximum, average and danger levels 46

3.5 Economic Order Quantity and quantity discounts 48

3.6 ABC, FSN, HML, VED, JIT and perpetual inventory 50

3.7 Inventory turnover ratios and stock discrepancies 53

3.8 Waste, scrap, spoilage, defectives and obsolescence 55

4 Employee Cost and Direct Expenses 59

4.1 Attendance, payroll and the elements of wages 59

4.2 Idle time and overtime 62

4.3 Employee turnover: measurement, causes, effects and cost 64

4.4 Time rate, piece rate and the Halsey and Rowan premium plans 67

4.5 Direct expenses 70

5 Overheads: Absorption Costing 73

5.1 Classifying overheads, allocation and apportionment 73

5.2 Primary distribution of factory overheads 75

5.3 Secondary distribution: direct, step, repeated and simultaneous methods 77

5.4 Absorption rates and the machine hour rate 80

5.5 Under- and over-absorption of overheads 83

5.6 Administration, selling, distribution and R&D overheads 85

6 Activity Based Costing 89

6.1 Activities, cost pools, cost drivers and the cost hierarchy 89

6.2 Product costs under ABC versus traditional absorption 91

6.3 Activity based management and budgeting; benefits and limits 94

7 Cost Accounting Systems 97

7.1 Non-integrated and integrated accounting systems 97

7.2 Reconciling cost and financial profit 99

8 Unit and Batch Costing 102

8.1 Unit costing 102

8.2 Batch costing and the economic batch quantity 104

9 Job Costing 107

9.1 Job costing 107

10 Process and Operation Costing 110

10.1 Process accounts, normal loss, abnormal loss and abnormal gain 110

10.2 Equivalent units: FIFO and weighted average 112

10.3 Inter-process profit and operation costing 115

11 Joint Products and By-Products 118

11.1 Apportioning joint costs: physical units, sales value and net realisable value 118

11.2 Sell at split-off or process further 121

11.3 Treatment of by-products 123

12 Service Costing 126

12.1 Service costing and composite cost units 126

12.2 Transport costing: passenger-km and tonne-km 128

12.3 Hotels, hospitals, power, IT and other services 130

13 Standard Costing 133

13.1 Standards, their types and standard costing 133

13.2 Material variances 136

13.3 Labour variances, including idle time 138

13.4 Variable overhead variances 140

13.5 Fixed overhead variances 142

13.6 Reconciling budgeted and actual profit, and accounting for variances 144

14 Marginal Costing 148

14.1 Marginal cost, contribution and the P/V ratio 148

14.2 Break-even point, margin of safety and angle of incidence 150

14.3 Break-even and profit-volume charts 153

14.4 Marginal versus absorption costing and profit reconciliation 155

14.5 Limiting factor and product mix 158

14.6 Make or buy, shut down and other short-term decisions 160

14.7 Multi-product break-even analysis 163

15 Budgets and Budgetary Control 165

15.1 Budgets, budgetary control, the budget manual and key factor 165

15.2 Functional budgets and the master budget 168

15.3 Flexible budgets 171

15.4 Cash budget 174

15.5 Zero based budgeting and performance budgeting 176

15.6 Control ratios, budget variances, motivation, feedback and feed-forward 179

CHAPTER 1

Introduction to Cost and Management Accounting

1.1 Objectives, scope and users of cost and management accounting

1 **ICAI's definition of costing?**

A The technique and process of ascertaining costs.

2 **ICAI's definition of cost (as a noun)?**

A The amount of expenditure (actual or notional) incurred on or attributable to a specified article, product or activity.

3 **ICAI's definition of cost accounting?**

A The process of accounting for cost which begins with the recording of income and expenditure or the bases on which they are calculated and ends with the preparation of periodical statements and reports for ascertaining and controlling costs.

4 **ICAI's definition of cost accountancy?**

A The application of costing and cost accounting principles, methods and techniques to the science, art and practice of cost control and the ascertainment of profitability, including the presentation of information derived from it for managerial decision making.

5 **Management accounting, as per CIMA Official Terminology?**

A The application of the principles of accounting and financial management to create, protect, preserve and increase value for the stakeholders of for-profit and not-for-profit enterprises in the public and private sectors.

6 **The five objectives of cost and management accounting, in ICAI's order?**

A Ascertainment of cost; determination of selling price and profitability; cost control; cost reduction; assisting management in decision making.

7 **The four steps of cost control?**

A (a) Determination of pre-determined standard or results; (b) measurement of actual performance; (c) comparison of actual performance with the set standard or target; (d) analysis of variance and action.

8 **The three-fold assumption in the definition of cost reduction?**

A There is a saving in unit cost; the saving is permanent; the utility and quality of the goods and services remain unaffected, if not improved.

9 **The seven functions in the scope of cost accounting?**

A Costing, cost accounting, cost analysis, cost comparisons, cost control, cost reports, statutory compliances.

10 **Which function does ICAI call the ultimate function of cost accounting?**

A Cost reports: prepared for management at different levels for planning and control, performance appraisal and managerial decision making.

11 **The internal and external users of cost and management accounting information?**

A Internal: policy makers, managers, operational level staff, employees. External: regulatory authorities, auditors, shareholders, creditors and lenders.

12 **How are regulatory authorities concerned with cost accounting information?**

A They use it for tariff determination, providing subsidies, rate fixation etc., and require it on the basis of set standards and formats.

1.2 Cost accounting department and installing a costing system

13 **The nine factors to be studied before installing a cost accounting system?**

A Objective; nature of business or industry; organisational hierarchy; knowing the product; knowing the production process; information synchronisation; method of maintenance of cost records; statutory compliances and audit; information attributes.

14 **Installing a costing system: what does the nature of the product decide?**

A The type of costing system: a product with by-products needs a system that accounts for by-products; for perishable or short shelf-life products, marginal costing is appropriate to know the contribution and minimum price.

15 **Organisational hierarchy: what information does each level of management need from the costing system?**

A Top management: corporate strategy. Strategic level: marketing strategy, product diversification, product pricing. Operational level: standard quantity to be consumed, report on idle time.

16 **The six essential features of a good cost accounting system?**

A Informative and simple; accurate and authentic; uniformity and consistency; integrated and inclusive; flexible and adaptive; trust on the system.

17 **Why must a cost accounting system be uniform and consistent?**

A Uniformity and consistency in classification, treatment and reporting of cost data are required for benchmarking and comparability of results, in both horizontal and vertical analysis.

18 **The three limitations of cost accounting?**

A It is expensive; reconciliation with financial accounts is required; work is duplicated because two sets of accounts are kept.

19 **The five functions of cost and management accounting?**

A Collection and accumulation of cost by element; assigning costs to cost objects; setting budgets and standards, comparing and reporting deviations; provision of relevant information for decision making (MIS); measuring and evaluating the performance of responsibility centres.

20 **Which is the main function of cost and management accounting?**

A Provision of relevant information to management for decision making, through a management information system (MIS).

21 **Practical difficulties in installing a costing system (ICAI RTP/MTP answers)?**

A Lack of top management support; resistance from cost accounting departmental staff; non-co-operation from user departments; shortage of trained staff.

22 **How are the practical difficulties of installing a costing system overcome?**

A Sell the idea to top management; a behavioural approach to resistance and non-co-operation; proper training at each level; regular meetings with cost staff, user departments and top management.

1.3 Cost, financial and management accounting, and financial management

23 **Cost accounting vs management accounting: the six bases of difference, in ICAI's order?**

A Nature, objective, area, recording of data, development, rules and regulation.

24 **Cost vs management accounting on nature?**

A Cost accounting records the quantitative aspect only; management accounting records both qualitative and quantitative aspects.

25 **Cost vs management accounting on area?**

A Cost accounting deals only with cost ascertainment; management accounting is wider in scope, including financial accounting, budgeting, taxation, planning etc.

26 **Cost vs management accounting on recording of data?**

A Cost accounting uses both past and present figures; management accounting is focused on the projection of figures for the future.

27 **Financial vs cost accounting: the seven bases of difference, in ICAI's order?**

A Objective, nature, recording of data, users of information, analysis of cost and profit, time period, presentation of information.

28 **Financial vs cost accounting on recording of data?**

A Financial accounting records historical data; cost accounting uses both historical and pre-determined costs.

29 **Financial vs cost accounting on analysis of cost and profit?**

A Financial accounting shows profit or loss segment-wise or as a whole; cost accounting gives cost details for each cost object (product, process, job, operation, contract etc.).

30 **Financial vs cost accounting on time period and presentation?**

A Financial statements are prepared usually for a year, in a set format; cost reports are prepared as and when required, generally with no set format.

31 **How does cost accounting feed financial accounting?**

A It accumulates and ascertains costs for goods sold and inventories, and provides inputs to record costs in the financial accounting system.

32 **ICAI's one-line relationship between cost and management accounting and financial management?**

A Cost and Management Accounting is an application of Financial Management for decision making purposes.

1.4 Cost units, cost centres, cost

objects and responsibility centres

33 **ICAI's definition of a cost object?**

A Anything for which a separate measurement of cost is required; it may be a product, a service, a project, a customer, a brand category, an activity, a department or a programme etc.

34 **ICAI's definition of a cost unit?**

A A unit of product, service or time (or combination of these) in relation to which costs may be ascertained or expressed.

35 **Cost unit: brick-making, brewing, hospitals, education?**

A Brick-making: 1,000 bricks. Brewing: barrel. Hospitals: patient day. Education: course, enrolled student, successful student.

36 **Cost unit: power, steel, transport, gas, cement?**

A Power: kilowatt-hour (kWh). Steel: ton. Transport: passenger-kilometre. Gas: cubic feet. Cement: ton or per bag.

37 **What is a cost driver? Give ICAI's examples.**

A A factor or variable which affects the level of cost, generally an activity responsible for cost incurrence; e.g. number of machine set-ups, number of purchase orders, hours spent on product inspection, number of tests performed.

38 **What are responsibility centres?**

A Departments or persons to whom management delegates responsibility and authority, held responsible for performance in terms of expenditure, revenue, profitability and return on investment.

39 **The four types of responsibility centres and how each is measured?**

A Cost centre: costs against standards or budgets. Revenue centre: revenue generation. Profit centre: profitability. Investment centre: return on investment besides profit.

40 **Standard cost centre vs discretionary cost centre?**

A Standard: output measurable, inputs specified, input-output ratio clearly identifiable, manager responsible for adverse variances. Discretionary: output not measurable in financial terms, no input-output ratio, inputs compared with the allocated budget (R&D, advertisement).

41 **What makes a responsibility centre an investment centre rather than a profit centre?**

A The authority to make capital investment decisions, with performance measured on return on investment (ROI) besides profit.

42 **ICAI's examples of a profit centre and of an investment centre?**

A Profit centre: decentralised branches of an organisation. Investment centre: Maharatna, Navratna and Miniratna companies of the Public Sector Undertakings of the Central Government.

43 **The two main types of cost centre when a department is sub-divided for cost collection?**

A Machine and personnel, depending on whether the work at the centre is done by machine or by people (SM 4.12).

1.5 Elements and classification of cost

44 ICAI's six bases of classification of costs, in order?

A By nature or element; by functions; by variability or behaviour; by controllability; by normality; by costs used in managerial decision making.

45 What are overheads, as ICAI defines them?

A The aggregate of indirect material costs, indirect labour costs and indirect expenses, subdivided into production (works), administration, selling and distribution overheads.

46 When is a direct material treated as indirect?

A When it is used in small quantities and identifying the quantity is not economically feasible; materials used for purposes ancillary to the business are also indirect.

47 ICAI's five functions for functional classification of cost?

A Production or manufacturing cost, administration cost, selling cost, distribution cost, and research and development cost etc.

48 **Controllable cost: ICAI's meaning?**

A Cost that can be controlled, typically by a cost, profit or investment centre manager; it can be influenced by the action of the manager heading the responsibility centre in which it is incurred.

49 **Abnormal cost: meaning and treatment?**

A Cost not normally incurred at a given level of output in the conditions in which that level is normally attained; charged to Costing Profit and Loss Account.

50 **Opportunity cost: ICAI's meaning and example?**

A The value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action; e.g. interest lost on a bank deposit withdrawn to finance expansion.

51 **Sunk cost: meaning and example?**

A Historical costs incurred in the past; they play no role in current decisions; e.g. the written-down value of a machine being considered for replacement.

52 **Shut down costs?**

A Costs which continue to be incurred even when a plant is temporarily shut down, e.g. rent, rates, depreciation: fixed costs that cannot be avoided during the temporary closure.

53 **Out-of-pocket cost?**

A The portion of total cost which involves cash outflow; a short-run concept used in decisions such as pricing in recession or make or buy; avoidable if the proposal is not accepted.

54 **Imputed cost?**

A A notional cost not involving any cash outlay, e.g. interest on capital not actually paid; similar to opportunity cost.

55 **Product cost under marginal costing and under absorption costing?**

A Marginal costing: variable manufacturing costs. Absorption costing: total manufacturing costs, variable and fixed.

1.6 Cost behaviour: fixed, variable,

semi-variable and step costs

56 **ICAI's definition of fixed costs?**

A Costs which are incurred for a period, and which, within certain output and turnover limits, tend to be unaffected by fluctuations in the levels of activity (output or turnover); they do not tend to increase or decrease with changes in output.

57 **ICAI's definition of semi-variable costs, with examples?**

A Costs which contain both fixed and variable components and are thus partly affected by fluctuations in the level of activity; e.g. telephone bills, gas and electricity.

58 **The five methods of segregating semi-variable costs, in ICAI's order?**

A Graphical method; high-low method; analytical method; comparison by period or level of activity method; least squares method.

59 **High-low method: how is the variable cost found?**

A Difference between the total cost at the highest and lowest volume ÷ difference between the sales value at the highest and lowest volume: the rate of variable cost in relation to sales value (with units or hours as volume, a rate per unit or hour).

60 **Graphical method: where is the fixed cost read?**

A Where the line of best fit through the plotted observations cuts the Y-axis (total cost); a line from there parallel to the X-axis shows the fixed cost.

61 **Analytical method?**

A An experienced cost accountant judges empirically the degree of variability of each semi-variable expense (e.g. 70% variable); the fixed part stays constant and the variable part changes with production.

62 **Comparison by period or level of activity: formula?**

A Variable cost per unit = change in the amount of expense ÷ change in the quantity of output; fixed = expense − variable cost at either level.

63 **Least squares method: the equation and its symbols?**

A y = mx + c: y total cost, x volume of output, m variable element of cost per unit, c total fixed cost. ICAI calls it the best method.

64 **What is a step cost, and how is it costed?**

A A cost fixed over a band of activity that jumps when activity crosses into the next band (e.g. one supervisor for every 20 workers); number of steps = activity ÷ band, rounded up, × cost per step.

65 **Primary objective of segregating semi-variable costs into fixed and variable?**

A To ascertain marginal costs; it also helps in controlling expenses, preparing flexible budgets and decision making.

1.7 Cost control and cost reduction

66 **ICAI's definition of cost reduction?**

A The achievement of real and permanent reduction in the unit cost of goods manufactured or services rendered without impairing their suitability for the use intended or diminution in the quality of the product.

67 **The three-fold assumptions in the definition of cost reduction?**

A (a) There is a saving in unit cost; (b) such saving is of permanent nature; (c) the utility and quality of the goods and services remain unaffected, if not improved.

68 **The four steps of cost control, in ICAI's order?**

A Determination of pre-determined standard or results; measurement of actual performance; comparison of actual performance with set standard or target; analysis of variance and action.

69 **The two actions taken for cost reduction?**

A Value analysis (segmenting activities into value added and non-value added and eliminating the non-value added, e.g. by Value Chain Analysis) and continuous research and study to find the most optimal way to produce or serve.

70 **Distinguish between cost control and cost reduction: ICAI's five points, in order?**

A (1) Maintains costs per established standards / challenges all standards and improvises them continuously; (2) lowest possible cost under existing conditions / no condition permanent; (3) emphasis on past and present / on present and future; (4) preventive function / corrective function, operating even with efficient cost control; (5) ends when targets are achieved / no visible end, a continuous process.

71 **Tapri's cost reduction cell has met this year's target of 40 paise a cup, and Kesar wants to wind it up. Which point of ICAI's table answers him?**

A Cost reduction has no visible end and is a continuous process; it is cost control that ends when targets are achieved.

72 **A thinner kulhad saves 5 paise a cup but breaks in customers' hands. Which of the three-fold assumptions does it fail?**

A The third: the utility and quality of the goods and services must remain unaffected, if not improved. So the saving is not cost reduction.

73 **ICAI's meaning of cost control, as an objective of cost accounting?**

A Maintaining discipline in expenditure: it ensures that expenditures are in consonance with predetermined set standards and that any variation from these standards is noted and reported on a continuous basis.

74 **Who developed Value Chain Analysis, named by ICAI as a method of value analysis for cost reduction?**

A Michael Porter.

1.8 Methods and techniques of costing, and digital costing

75 The methods of costing, in ICAI's order?

A Single or output costing; batch costing; job costing; contract costing; process costing (operation costing); operating costing; multiple costing.

76 The techniques of costing ICAI lists?

A Uniform costing; marginal costing; standard costing and variance analysis; historical costing (post costing, continuous costing); absorption costing.

77 Multiple costing: meaning and ICAI's example?

A A combination of two or more methods of costing; e.g. bicycle parts costed by job or batch costing and the assembly of the bicycle by single or output costing.

78 Operating costing is used for which concerns?

A Concerns rendering services, like transport, supply of water and retail trade.

79 **Post costing vs continuous costing?**

A Both are historical costing. Post costing: cost ascertained after production is completed. Continuous costing: cost ascertained as soon as the job is completed or even while the job is in progress.

80 **The two advantages of uniform costing?**

A (i) The performance of each firm can be compared with another's or with the industry average; (ii) the cost of production true for the industry as a whole can be determined, useful when tax relief or protection is sought from the Government.

81 **How does a digital costing system differ from the conventional cost accounting system?**

A It too collects, classifies and accounts for data to process information for decisions; the difference is the method of collection, medium of storage, forms of analysis and reporting, and it links production, procurement and inventory management with the systems of suppliers, customers and the market.

82 **The six kinds of data a digital costing system provides?**

A Cost incurred on a cost object; time spent; resource consumption; current market price of final product and raw materials; lead time and availability of materials; product demand and trend.

83 **Name five benefits of a digital costing system.**

A Any five of: cost with certainty on a cost object; time data for incentive plans; material requirement planning and JIT; elimination of non-value-added activities; real-time standards and variances; Marked to Market standards; demand prediction for budgets; better cost behaviour and cost-benefit analysis.

84 **Impact of IT on cost accounting: what does ERP integration achieve?**

A A single entry into the accounting system provides custom-made reports for every purpose, saves preparing different sets of documents, and makes reconciliation of cost and financial accounts simpler and less cumbersome.

CHAPTER 2

Cost Sheet

2.1 Cost sheet for a manufacturer

1 **ICAI's functional classification of costs, in order?**

A (i) Production / manufacturing cost, (ii) administration cost, (iii) selling cost, (iv) distribution cost, (v) research and development costs.

2 **The four cost heads in a cost sheet?**

A Prime cost, cost of production, cost of goods sold, cost of sales.

3 **Define prime cost.**

A The total of direct materials costs, direct employee (labour) costs and direct expenses.

4 **What is added to works cost to arrive at cost of production?**

A Quality control cost, R&D cost (process related), administrative overheads related to production and primary packing, less credit for recoveries (scrap, waste, by-products).

5 **Primary versus secondary packing: where does each go?**

A Primary packing (holds and preserves the product) is in cost of production; secondary packing (to store, transport and promote) is in cost of sales with selling and distribution.

6 **Treatment of interest and finance charges in a cost sheet?**

A Not included in cost of production; presented as a separate item of cost of sales.

7 **Treatment of abnormal costs in a cost sheet?**

A Where material and quantifiable, they do not form part of cost of production or acquisition or supply of goods or provision of service (e.g. costs of a pandemic, employee costs of a sudden lockdown).

8 **Treatment of a subsidy, grant or incentive?**

A Reduced from the cost object to which the amount pertains.

9 **Treatment of penalty, fine, damages and demurrage?**

A They do not form part of cost.

10 **When is GST on raw materials part of material cost?**

A Only when input tax credit is not available or not availed.

11 **Advantages of a cost sheet (SM 6.10)?**

A Total cost and cost per unit; cost comparison; cost estimates for tenders; arriving at selling price; cost control by disclosing operational efficiency.

12 **Units produced from units sold and stocks?**

A Units produced = units sold + closing stock − opening stock.

2.2 Estimated cost sheets, tenders and quotations

13 **Kohler's definition of estimated cost?**

A The expected cost of manufacture, or acquisition, often in terms of a unit of product, computed on the basis of information available in advance of actual production or purchase.

14 **Why are production overheads pre-determined (SM 4.41)?**

A Because they are to be determined in advance for fixing selling price, quoting tender price and formulating budgets.

15 **Formula: pre-determined overhead rate?**

A Estimated or normal overheads for the period ÷ budgeted number of units (or hours, or wages) during the period.

16 **Usual recovery bases in a quotation question?**

A Factory overheads as a % of direct wages (or prime cost or hours); administration as a % of works cost; selling and distribution as a % of cost of production.

17 **Profit of 20% on selling price is what % on cost?**

A 25% (x ÷ (100 – x) = 20 ÷ 80).

18 **Total cost from selling price and profit % on cost?**

A Total cost = selling price ÷ (100% + profit %).

19 **How are costs specific to one order (delivery, a special die) treated in a quotation?**

A Added at their own estimated amounts, not through a recovery rate.

20 **Two past jobs give selling prices and profit % on cost. How do you find the factory and administration rates?**

A Find each job's total cost, write (materials + wages + F × wages)(1 + A) = total cost for each, and solve the two equations for F and A.

21 **At a higher volume, what happens to fixed cost per unit and variable cost per unit?**

A Fixed cost per unit falls (total unchanged); variable cost per unit stays the same.

22 **Which advantage of a cost sheet does a quotation use?**

A It facilitates the preparation of cost estimates required for submitting tenders (and helps in arriving at the selling price).

2.3 Cost sheet for the service sector

23 ICAI's four differences between service costing and product costing?

A Tangibility (services cannot be stored, no inventory); composite cost units; employee cost is a major element over material; indirect costs such as administration are significant and tracing may not be economically feasible.

24 On what bases may a service cost statement be prepared?

A On functional classification (as for products) or on variability.

25 The three heads of a service cost statement on variability?

A Fixed costs or standing charges; variable costs or operating (running) expenses; semi-variable costs or maintenance expenses.

26 What if no information on semi-variable costs is given?

A The costs are shown under fixed and variable heads only.

27 **Treatment of depreciation in a service cost statement?**

A Fixed if related to effluxion of time or calculated on a time basis; variable if calculated on the basis of activity level or usage.

28 **Treatment of interest in a service cost statement?**

A Presented as a separate item of cost of sales; in general treated as a fixed cost unless otherwise given.

29 **Service costing is also known as?**

A Operating costing.

30 **Internal and external applications of service costing?**

A Internal: in-house support services (canteen, staff hospital, boiler house, captive power, vehicle fleet, IT department). External: services offered to outside customers as a profit centre (transport, hotels, financial services, insurance, IT).

31 **Takings when commission is c% and profit p% of takings?**

A Takings = total cost ÷ (1 − c − p).

32 **Cost unit examples for canteen, hospital, hotel, electricity, IT?**

A Canteen: per item or meal; hospital: per patient-day, bed or operation; hotel: guest-days or room-days; electricity: kWh; IT: per project or module.

CHAPTER 3

Material Cost

3.1 Procurement procedure, GeM, ERP and stock verification

1 **ICAI's definition of a bill of materials?**

A A detailed list specifying the standard quantities and qualities of materials and components required for producing a product or carrying out of any job; also called materials specification list. Prepared by the engineering or planning department.

2 **What is a material requisition note, and who prepares it?**

A A voucher of authority used to get materials issued from store (material requisition slip). Generally prepared by the production or other consuming department.

3 **Who fills up the purchase requisition?**

A The store keeper for regular materials; the departmental head for special materials not stocked as regular items.

4 **ICAI's definition of a purchase order?**

A A written request to the supplier to supply specified materials at specified rates and within a specified period.

5 **GeM: what is it, what tools does it provide, and is it optional for government users?**

A Government e-Marketplace, a dedicated e-market for goods and services procured by government organisations, departments and PSUs; e-bidding, reverse e-auction and demand aggregation. Purchases through GeM by government users are authorised and made mandatory by the Ministry of Finance.

6 **How many copies of a goods received note, and who gets them?**

A Generally four: purchase department, store or order indenting department, receiving department and accounting department.

7 **Material returned to the supplier before the receiving report is prepared: is a stores ledger adjustment needed?**

A No: it is not in the receiving report or the stores ledger. A return after the entry needs a material returned note to exclude its value from the stores account.

8 **Copies of a material transfer note and of a shop credit note?**

A Transfer note: duplicate, to the cost accounting department and the department making the transfer (no store copy). Shop credit note: triplicate, to the store room, the cost accounting department and the department returning it.

9 **Bin card vs stores ledger: who maintains each, and what does each show?**

A Bin card: store keeper, in the store; quantities only. Stores ledger: cost accounting department; quantity and value.

10 **Against which documents is a supplier's invoice checked before it is passed?**

A Quantity against the goods received note; price against the purchase order.

11 **Continuous stock verification: by whom, how often, with what notice?**

A By staff independent of the store and production staff (for example internal audit), at appropriate intervals, without prior notice; the element of surprise is essential.

12 **Advantages of continuous stock-taking (five)?**

A No closure of normal functioning; discrepancies noticed and corrected earlier; sobering influence on stores staff; closer watch on movement, so less obsolescence buying; final accounts ready quickly and interim accounts possible.

3.2 Valuing material receipts

13 **ICAI's definition of valuation of material receipts?**

A Ascertainment of cost of material purchased; cost includes cost of purchase net of trade discounts, rebates, duty draw-back, input credit availed etc., and other costs incurred in bringing the inventories to their present location and condition.

14 **Treatment of cash discount in valuing material receipts?**

A Not deducted from the purchase price; treated as interest and finance item and ignored.

15 **Treatment of GST in valuing material receipts?**

A Excluded from the cost of purchase if credit for it is available; unless mentioned specifically, it should not form part of cost of purchase.

16 **Treatment of demurrage?**

A A penalty imposed by the transporter for delay in loading or unloading: an abnormal cost, not included in the cost of purchase.

17 **Treatment of detention charges or fines?**

A Imposed for non-compliance of rule or law by a statutory authority: abnormal cost, not included in the cost of purchase.

18 **Treatment of returnable containers?**

A If returned and refunded, not considered; if the refund is less than the amount paid, only the shortfall is added to the cost of purchase. Non-returnable containers are added.

19 **Treatment of a subsidy, grant or incentive received on purchase of materials?**

A Deducted from the cost of purchase.

20 **Shortage due to normal vs abnormal reasons in material receipts?**

A Normal (breaking of bulk, evaporation, unavoidable): absorbed by the good units. Abnormal (mishandling, pilferage, avoidable): not absorbed; debited to the Costing Profit and Loss Account.

21 **Formula: cost per unit of material received with a normal shortage?**

A Total cost of material ÷ (units purchased − normal shortage).

22 **"Other expenses at 2% of total cost": how are they calculated?**

A Cost before other expenses × 2/98, because the total cost includes the other expenses.

3.3 Pricing material issues: FIFO, LIFO and average methods

23 **FIFO: how are issues priced and at what price is closing stock left?**

A Issues are priced in the order in which the materials were purchased (oldest first); the closing stock is at the price of the latest purchases.

24 **LIFO: how are issues priced, and is it permitted under AS 2?**

A The last lot purchased is issued first until exhausted, then the earlier lot; closing stock is at the oldest prices. LIFO is not permitted under AS 2 or Ind AS 2.

25 **Formula: weighted average price, and when is it recomputed?**

A Total cost of material in stock ÷ total quantity in stock; recomputed every time a new lot is received.

26 **Simple average price method: formula and when suitable?**

A Total of the unit prices of each purchase ÷ number of purchases (quantities ignored; fully issued lots left out). Suitable when lots are of uniform quantity and prices do not fluctuate considerably.

27 **Which method is considered suitable in times of falling prices, and why?**

A FIFO: the material cost charged to production is high while the replacement cost of materials is low.

28 **ICAI's four groups of methods of pricing material issues?**

A Cost price (specific, FIFO, LIFO, base stock); average price (simple, weighted); market price (replacement, realisable); notional price (standard, inflated, re-use).

29 **Five factors for selecting a method of pricing issues?**

A Frequency of purchases and price fluctuations; frequency and quantity of issues; nature of the cost accounting system; nature of business and production process; management policy on closing stock valuation.

30 **Material returned to stores by a department: how is it priced?**

A Entered in the receipt column at the price at which it was originally issued.

31 **Shortage found during physical verification: how is it entered in the stores ledger?**

A In the issue column, valued at the rate of the method adopted (FIFO or other). An abnormal loss goes to Costing P&L; a normal loss is absorbed by the good units.

32 **Transfer of material from one job to another: stores ledger entry?**

A None; the material does not pass through the store. Only the cost accounts of the jobs are adjusted (material transfer note).

33 **In rising prices, rank FIFO, LIFO and weighted average by cost of materials issued.**

A FIFO lowest, weighted average in between, LIFO highest (closing stock in the reverse order).

3.4 Stock levels: re-order, minimum, maximum, average and danger levels

34 **Formula: re-order level (two ways)?**

A Maximum consumption × maximum re-order period; or minimum stock level + (average consumption × average re-order period).

35 **Formula: minimum stock level?**

A Re-order level − (average consumption × average re-order period).

36 **Formula: maximum stock level?**

A Re-order level + re-order quantity − (minimum consumption × minimum re-order period). The re-order quantity may be the EOQ.

37 **Formula: average stock level (two ways)?**

A Minimum stock level + ½ re-order quantity; or (maximum level + minimum level) ÷ 2.

38 **Danger level: definition and formula?**

A The level at which normal issues of raw material are stopped and only emergency issues are made. Average consumption (sometimes minimum) × lead time for emergency purchase.

39 **ICAI's definition of minimum stock level?**

A The lowest level of material stock which must be maintained in hand at all times, so that there is no stoppage of production due to non-availability of inventory.

40 **What is lead time (re-order period)?**

A The time required to get an order from the supplier to the stores; on the stock control chart, the interval between placing the order at the re-order level and receiving the material.

41 **Re-order level vs re-order quantity?**

A Re-order level: the stock level at which a fresh order is placed (when to order). Re-order quantity: the quantity ordered (how much), ideally the EOQ.

42 **Minimum consumption is not given, but average and maximum are. How is it found?**

A Average = (maximum + minimum) ÷ 2, so minimum = 2 × average − maximum.

43 **Maximum stock level of a perishable material?**

A The lower of the formula figure (ROL + ROQ − min. consumption × min. lead time) and the consumption during its usable life.

44 **Safety stock: what is it, and how is its optimum level found?**

A Stock maintained in excess of lead time consumption as a cushion against unexpected demand; the optimum is where the total of expected stock-out cost and carrying cost is least.

3.5 Economic Order Quantity and quantity discounts

45 **Formula: Economic Order Quantity?**

A EOQ = $\sqrt{(2AO \div C)}$, where A = annual requirement, O = cost per order, C = carrying cost per unit per annum.

46 **ICAI's definition of Economic Order Quantity?**

A The size of an order for which total of ordering and carrying cost are minimum.

47 **Give ICAI's examples of ordering costs.**

A Cost to invite quotations, documentation such as preparing purchase orders, employee cost directly attributable to procurement, transportation and inspection cost.

48 **Give ICAI's examples of carrying costs.**

A Cost of fund invested in inventories, cost of storage, insurance cost and obsolescence.

49 **The four assumptions underlying EOQ, in ICAI's order?**

A (i) Ordering cost per order and carrying cost per unit per annum are known and fixed; (ii) anticipated usage in units is known; (iii) cost per unit of material is constant and known; (iv) the quantity ordered is received immediately (lead time zero).

50 **At the EOQ, how do annual ordering cost and annual carrying cost compare?**

A They are equal, and their sum equals EOQ × carrying cost per unit per annum (basic model).

51 **How is a quantity discount offer evaluated?**

A Compare total annual cost (purchase cost + ordering cost + carrying cost) at the EOQ with that at the discount order size, carrying cost on the discounted price; accept if the total is lower.

52 **Storage 2%, interest 12% and obsolescence 6% per annum on a ₹20 unit: carrying cost per unit per annum?**

A 20% × ₹20 = ₹4 per unit per annum (SM Illustration 4).

53 **With several price breaks, which order sizes are tested?**

A When the EOQ at the basic price falls in the undiscounted band: the EOQ and the lowest order quantity of each discount band; the one with the lowest total cost is the most economical purchase level.

54 **Carrying cost is a % of price. The price is multiplied by 4: what happens to the EOQ?**

A It halves, because EOQ moves with $1 \div \sqrt{C}$.

3.6 ABC, FSN, HML, VED,

JIT and perpetual inventory

55 **ICAI (CIMA): define inventory control.**

A The function of ensuring that sufficient goods are retained in stock to meet all requirements without carrying unnecessarily large stocks.

56 **The four bases of inventory control?**

A By setting quantitative levels; on the basis of relative classification; using ratio analysis; physical control.

57 **ABC analysis: typical share of items and value in each category?**

A A: about 10% of items, about 70% of value. B: about 20% and 20%. C: about 70% of items, about 10% of value.

58 **The four advantages of ABC analysis?**

A Continuity in production; lower cost; less attention required; systematic working.

59 **Basis of FSN, VED and HML classification?**

A FSN: frequency of usage (inventory turnover). VED: criticality for the production function and final product. HML: cost of an individual item.

60 **VED: how is each category controlled?**

A Vital: strictly controlled by setting a re-order level. Essential: reviewed periodically, second priority. Desirable: optional in nature; unavailability causes no production or efficiency loss.

61 **How are non-moving items treated under FSN?**

A Kept for disposal, reported to management, and an appropriate provision for loss may be created.

62 **JIT: its other names and its two principles?**

A 'Demand pull' or 'Pull through' system. (i) Produce goods only when it is required; (ii) deliver products to customers only when they want them.

63 **Two bin system: when is a fresh order placed?**

A As soon as issues have to be made out of the smaller part of the bin, which holds a quantity equal to the minimum stock or the re-order level.

64 **Perpetual inventory comprises what, and its success needs which final three steps?**

A Bin cards and the stores ledger. Explain promptly the causes of discrepancies; make corrective entries; remove the causes.

65 **How is slow-moving and non-moving stock detected (ICAI's four ways)?**

A Periodic status reports; inventory turnover period in days or months of consumption; inventory turnover ratio computed periodically; a well-designed information system.

66 **Five advantages of perpetual inventory?**

A Stocks counted and book balances adjusted whenever desired; quick interim Profit and Loss Account; discrepancies easily located and corrected; surplus, dormant, obsolete and slow-moving materials revealed in time; stocks kept within the levels and purchase requisitions raised for the correct quantity at the right time.

3.7 Inventory turnover ratios and stock discrepancies

67 **Formula: inventory turnover ratio (Paper 4)?**

A Cost of materials consumed during the period ÷ cost of average stock held during the period; average stock = ½ (opening + closing stock).

68 **Formula: average number of days of inventory holding?**

A 365 days (or 12 months) ÷ inventory turnover ratio.

69 **What do a high and a low inventory turnover ratio indicate?**

A High: the material is fast moving. Low: over-investment and locking up of working capital in inventories.

70 **Define the input-output ratio.**

A The ratio of the quantity of input of material to production and the standard material content of the actual output; it shows whether usage is favourable or adverse.

71 **How is material found short on physical verification valued?**

A Entered in the issue column at the rate as per the issue pricing method adopted (FIFO or any other).

72 **Causes of normal and abnormal shortage of materials?**

A Normal: breaking of bulk, evaporation, unavoidable conditions. Abnormal: mishandling, pilferage, avoidable reasons.

73 **Treatment of a normal and an abnormal shortage in stores?**

A Normal: absorbed by good units (or charged to factory overheads). Abnormal: valued at the issue rate and transferred to Costing Profit and Loss Account.

74 **Journal entry for an abnormal stock shortage in the cost ledger?**

A Costing Profit and Loss A/c Dr; To Stores Ledger Control A/c.

75 **Journal entry for a normal stock deficiency in the cost ledger?**

A Factory Overhead Control A/c Dr; To Stores Ledger Control A/c.

76 **Material consumption in financial accounts?**

A Opening stock + purchases − closing stock (stock at cost or net realisable value, whichever is less).

3.8 Waste, scrap, spoilage,

defectives and obsolescence

77 **ICAI's definition of waste, and its treatment?**

A The portion of raw material lost during storage or production and discarded (it may or may not have value). Normal: absorbed by good units. Abnormal: transferred to Costing Profit and Loss Account.

78 **Treatment of normal and abnormal scrap?**

A Normal: cost borne by good units; realisable value deducted from the cost. Abnormal: scrap account charged with full cost, job or process credited; profit or loss on realisation to Costing Profit and Loss Account.

79 **Treatment of normal and abnormal spoilage?**

A Normal: included in costs, charged to the production order or to production overhead; value realised credited to it. Abnormal: charged to Costing Profit and Loss Account.

80 **Spoiled work caused by rigid specification: treatment?**

A Cost of spoiled work absorbed by good production; cost of disposal charged to production overhead.

81 **Treatment of normal and abnormal defectives?**

A Normal: cost less realisable value charged to material cost of good production. Abnormal: excluded from material cost, treated as a loss after crediting realisable value, transferred to Costing Profit and Loss Account.

82 **Reworks and rejects?**

A Reworks: defectives re-made to standard with additional material (repairs, reconditioning, refurbishing). Rejects: defectives that cannot reach the standard; disposed of or recycled.

83 **Define obsolescence and state its treatment.**

A The loss in the intrinsic value of an asset due to its supersession. Obsolete material is a total loss, disposed of at the best price; the loss is abnormal and not part of the cost of manufacture.

84 **Three circumstances in which materials become obsolete?**

A A spare part or component of a machine that is now obsolete; material for a product that has become obsolete; material replaced by another due to improved quality or a fall in price.

85 **Defect due to the Inspection Department wrongly accepting poor material: treatment?**

A Rectification charged to the department, not to the batch's cost of manufacture; written off to Costing Profit and Loss Account as abnormal.

86 **Waste against scrap: three differences?**

A Waste is connected with input, may be invisible, generally has no recoverable value. Scrap is connected with output, identifiable with physical substance, small recoverable value (termed by-products).

87 **The five steps of reclaiming loss from defective units?**

A Send to a fixed place; dismantle; take good parts into stock; send repairable parts to the workshop, then to stock; collect unusable parts to be melted or sold off.

CHAPTER 4

Employee Cost and Direct Expenses

4.1 Attendance, payroll and the elements of wages

1 **ICAI's definition of employee (labour) cost?**

A Benefits paid or payable to the employees of an entity, whether permanent or temporary, for the services rendered by them; it includes payments made in cash or kind.

2 **Difference between time-keeping and time booking?**

A Time-keeping records employees' attendance time (total time in the factory); time booking records the time spent by an employee on each job or process.

3 **Six objectives of time-keeping?**

A Preparation of payrolls; calculating overtime; ascertaining and controlling employee cost; ascertaining idle time; disciplinary purposes; overhead distribution.

4 **Three purposes of time booking?**

A Costing (cost of each job or activity), measurement of efficiency (actual vs standard time), fixation of responsibility (controllable and uncontrollable variances).

5 **Payroll procedure: the five steps in order?**

A Time and attendance details; list of employees and other details; computation of wages and other incentives; payment to employees after deductions; deposit of statutory liabilities.

6 **Four statutory deductions from the payroll?**

A Employee's contribution to provident fund; employee's contribution to ESI; tax deducted at source; professional tax.

7 **Bio-metric attendance system: how does it work and what does it prevent?**

A Attendance is marked by recognising physical and behavioural traits (fingerprint, face, retina) matched against a database; it reduces time manipulation and proxy attendance, but its cost may not suit small organisations.

8 **Formula: rate per hour for costing purposes (direct workers)?**

A (Total estimated monetary benefits + cost of non-monetary benefits) ÷ (budgeted direct employee hours – normal idle time).

9 **Five non-monetary benefits included in wages?**

A Medical facilities; educational and training facilities; recreational and sports facilities; housing and social welfare; cost of subsidised canteen and co-operative societies.

10 **Treatment of night shift allowance in cost accounts?**

A Treated as overheads, because the extra payment is not for any particular job.

11 **Two treatments of holiday and leave wages?**

A As departmental overheads (recorded separately from "worked for wages"), or by inflating the wage rate for costing, which is possible only for direct workers.

12 **Eight factors for the control of employee cost?**

A Assessment of manpower requirements; control over time-keeping and time-booking; time and motion study; control over idle time and overtime; control over employee turnover; wage and incentive systems; job evaluation and merit rating; employee productivity.

4.2 Idle time and overtime

13 **Define idle time.**

A The time during which no production is carried out because the worker remains idle but is paid; the difference between the time paid and the time booked.

14 **Four causes of normal idle time?**

A Time lost between factory gate and place of work; interval between one job and another; setting-up time for the machine; normal rest time and lunch break.

15 **Treatment of normal idle time in cost accounts?**

A Part of the cost of production: for direct workers an allowance in standard hours or rate; for indirect workers it is considered in the overhead rate.

16 **Treatment of abnormal idle time?**

A Not included in production cost; shown as a separate item in the Costing Profit and Loss Account, analysed into controllable and uncontrollable by cause.

17 **Controllable vs uncontrollable abnormal idle time?**

A Controllable: could have been put to productive use had management been more alert and efficient. Uncontrollable: lost to abnormal causes beyond management's control, such as machine breakdown or flood.

18 **Two elements of overtime payment?**

A Wages for overtime hours at the normal rate, and the premium (extra) payment for overtime work.

19 **Overtime premium at the desire of the customer: treatment?**

A Charged to the job directly.

20 **Overtime to meet general production programmes or urgent orders: treatment of the premium?**

A Overhead cost of the department or cost centre that works the overtime.

21 **Overtime due to the fault of another department, or due to flood or earthquake: treatment of the premium?**

A Fault of another department: charged to that department. Abnormal conditions: not charged to cost; Costing Profit and Loss Account.

22 **Overtime worked regularly as a policy (shortage of workers): how is the job charged?**

A At the average inflated wage rate (total wages including premium ÷ total hours), so the premium is part of employee cost.

23 **Factories Act, as quoted in the SM: when is overtime at twice the ordinary rate due?**

A When a worker works more than nine hours in any day or more than forty-eight hours in any week.

24 **What does persistent overtime indicate?**

A That the firm needs more capacity in men and machines, or that workers postpone ordinary work to the evening to earn overtime wages.

4.3 Employee turnover:

measurement, causes, effects and cost

25 **ICAI's definition of employee (labour) turnover?**

A The rate of change in the composition of employee force during a specified period measured against a suitable index.

26 **Formula: employee turnover by the replacement method?**

A Number of employees replaced during the period ÷ average number of employees on roll during the period × 100 (expansion hires excluded).

27 **Formula: employee turnover by the separation method?**

A Number of employees separated during the period ÷ average number of employees on roll during the period × 100.

28 **Formula: employee turnover by the flux method?**

A (Number separated + number replaced) ÷ average number on roll × 100; with accessions, (separations + accessions) ÷ average on roll × 100.

29 **What are accessions?**

A The total number of workers joining, including replacements.

30 **Formula: equivalent annual employee turnover rate?**

A Turnover rate for the period × 365 ÷ number of days in the period (a quarterly rate × 4).

31 **Three heads of causes of employee turnover?**

A Personal causes, unavoidable causes and avoidable causes.

32 **Five avoidable causes of employee turnover?**

A Dissatisfaction with job, remuneration, hours or conditions; strained relationship with management, supervisors or fellow workers; lack of training facilities and promotional avenues; lack of recreational and medical facilities; low wages and allowances.

33 **Five ways high employee turnover increases the cost of production?**

A Even flow of production disturbed; low efficiency of new workers; increased cost of training and induction; more breakage of tools and wastage of materials; increased cost of recruitment and training.

34 **Two types of cost associated with employee turnover?**

A Preventive costs (medical benefits, welfare like pension, other retention benefits) and replacement costs (recruitment, training and induction, abnormal breakage and scrap, extra wages and overheads from inefficient new workers).

35 **Profit forgone on account of employee turnover: how is the contribution lost computed?**

A Sales ÷ actual productive hours × productive hours lost (potential hours lost + unproductive training hours) × P/V ratio.

4.4 Time rate, piece rate and the Halsey and Rowan premium plans

36 **Formula: earnings under the Halsey premium plan?**

A Time taken × time rate + 50% of time saved × time rate.

37 **Formula: earnings under the Rowan premium plan?**

A Time taken × rate per hour + (time saved ÷ time allowed) × time taken × rate per hour.

38 **When do the Halsey (50%) and Rowan plans give the same bonus?**

A When the actual time taken is 50% of the time allowed.

39 **Time allowed 20 hours, time taken 15 hours, same hourly rate: which pays the worker more, the Halsey plan (50%) or the Rowan plan?**

A Rowan. Time taken is 75% of time allowed (more than 50%), so Rowan pays 75% of the time saved against Halsey's 50%: bonus 3.75 hours' wages against 2.5 (SM 3.33).

40 **At what time taken is the Rowan bonus at its maximum?**

A When the time taken is half the time allowed.

41 **Three cases where the time rate system is suitable?**

A Services that cannot be directly or tangibly measured (helpers, supervisory and clerical staff); highly skilled jobs; where the pace of output is independent of the operator (automatic chemical plants).

42 **Two advantages and two disadvantages of the Halsey plan?**

A Advantages: time rate guaranteed with a chance to earn more; equitable, as the employer gains from better methods and equipment. Disadvantages: incentive weaker than piece rate; the sharing principle may not be liked by employees.

43 **Three advantages of the Rowan plan?**

A Fool-proof: a worker can never double his earnings even with bad rate setting; suits moderately efficient workers better than Halsey; the sharing principle appeals to the employer as equitable.

44 **Two disadvantages of the Rowan plan beyond its sharing principle?**

A It is a bit complicated; the incentive is weak at high production where time saved exceeds 50% of time allowed.

45 **Formula: efficiency rating in per cent?**

A Time allowed as per standard ÷ time taken × 100.

46 **Premium bonus method: what is guaranteed and what is shared?**

A Day wages are guaranteed (except in the Barth system) when output is below or up to standard; the time saved is shared between employee and employer.

4.5 Direct expenses

47 **ICAI's definition of direct expenses?**

A Expenses other than direct material cost and direct employee cost, incurred to manufacture a product or for provision of service, which can be directly traced in an economically feasible manner to a cost object.

48 **ICAI's examples of direct expenses, in order?**

A Royalty paid or payable for production or provision of service; hire charges for hiring specific equipment; cost of product or service specific design or drawing; cost of product or service specific software; other expenses directly related with production or provision of service.

49 **Cost Sheet chapter (SM 6.4): three examples of direct expenses not in Chapter 3's list?**

A Cost of utilities such as power and fuel, steam etc.; fee for technical assistance and know-how; amortised cost of moulds, patterns, patents etc.

50 **How are direct expenses measured?**

A At invoice or agreed price net of rebate or discount, including duties and taxes for which input credit is not available, commission and other directly attributable costs.

51 **How is sub-contracted (job) work measured as a direct expense?**

A At the agreed price; where the principal supplies materials to the job workers, the value of those materials and other incidental expenses are added to the job charges.

52 **Treatment of a lump-sum or one-time direct expense?**

A Amortised over the estimated production volume or the benefit derived.

53 **Treatment of a direct expense of insignificant amount?**

A It may be treated as part of overheads.

54 **In which element of the cost sheet do direct expenses appear?**

A Prime cost (direct materials + direct employee cost + direct expenses).

55 **Salary of the marketing manager, office administration cost and sales commission: direct expenses?**

A No: salary of the marketing manager and sales commission are selling overheads; office and administration cost is an overhead.

CHAPTER 5

Overheads: Absorption Costing

5.1 Classifying overheads, allocation and apportionment

1 **ICAI's definition of overheads (SM 4.2)?**

A Expenditure which cannot be conveniently traced to or identified with any particular cost unit; incurred for output generally, not for a particular work order.

2 **The four classifications of overheads, in ICAI's order?**

A By function, by nature (behaviour), by element, by control.

3 **Define cost allocation.**

A The assignment or allotment of an entire item of cost to a particular cost centre or cost unit.

4 **Define cost apportionment.**

A The allotment of proportions of items of cost to cost centres or departments.

5 **What decides whether an expense is allocated or apportioned?**

A The relationship between the expense and the cost centre, not the nature of the expense.

6 **The two principles for assigning manufacturing overheads?**

A Cause and effect, and benefit received.

7 **Four sources for collecting factory overheads?**

A Invoices, stores requisitions, the wage analysis book and journal entries.

8 **What is a standing order?**

A A sanction for indirect expenses under various heads of expenditure; each head gets a number that continues from year to year.

9 **On what are variable and fixed manufacturing overheads absorbed?**

A Variable on actual production; fixed on normal capacity.

10 **Four advantages of departmentalisation?**

A Better estimation of expenses, better control, ascertainment of cost for each department, a suitable method of costing for each department.

11 **Primary, secondary and fancy packing: which cost does each belong to?**

A Primary packing: production cost. Packing for transport to the customer: distribution cost. Fancy packing: selling overhead.

5.2 Primary distribution of factory overheads

12 **What is primary distribution of overheads?**

A Allocating and apportioning factory overheads to all departments, production and service, so that every item sits in some department; the result is the primary distribution summary.

13 **Basis for rent, lighting and heating, fire precaution and air-conditioning?**

A Floor area, or volume of department.

14 **Basis for supervision, time keeping, labour welfare and the personnel office?**

A Number of workers.

15 **Basis for holiday pay, ESI and PF contribution and compensation to workers?**

A Direct wages.

16 **Basis for depreciation, and repairs and maintenance, of plant and machinery?**

A Capital values.

17 **Bases for electric power for machine operation?**

A Horse power of machines, number of machine hours, value of machines, or units consumed (H.P. × machine hours combines the first two).

18 **Basis for power or steam consumption, internal transport and managerial salaries?**

A Technical estimates.

19 **The three other bases of apportionment besides benefit received?**

A Analysis or survey of existing conditions; ability to pay; efficiency or incentive.

20 **Why may the ability-to-pay basis be inequitable?**

A The charge may have no relation to the actual effort involved, so easy-selling lines bear the largest proportion of expenses when they should bear the lowest.

21 **Four considerations in selecting a base?**

A Economy, practicability, equitability and reliability.

5.3 Secondary distribution: direct, step, repeated and simultaneous methods

22 **What is secondary distribution?**

A The re-apportionment of service department costs to production departments.

23 **The three methods of re-apportionment (SM 4.18)?**

A Direct re-distribution method; step method of secondary distribution (non-reciprocal method); reciprocal service method.

24 **The three reciprocal service methods?**

A Simultaneous equation method, trial and error method, repeated distribution method.

25 **Direct re-distribution method in one line?**

A Service department costs are apportioned over the production departments only, ignoring the services rendered by one service department to the other.

26 **Step method: which service department is apportioned first?**

A The one that renders services to the largest number of other departments; some authors would take the one with the largest cost first.

27 **Step method: what does a service department already closed receive?**

A Nothing; the last service department's cost goes to the production departments only.

28 **Trial and error method: how does it work?**

A The service departments' costs are apportioned back and forth between the service departments only until the amount is negligible; the totals are then apportioned to production departments. Same result as simultaneous equations.

29 **Repeated distribution method: how does it work?**

A Each service department's cost is distributed to other service and production departments on agreed percentages, repeatedly, until the service departments' figures are exhausted or too small.

30 **Basis for re-apportioning the stores department's cost?**

A Number of requisitions, or weight or value of materials issued.

31 **Bases for canteen, hospital and personnel departments?**

A Number of direct workers, number of employees.

32 **Bases for the maintenance and repair shop, planning and progress, and tool room?**

A Direct labour hours, machine hours, direct labour wages, asset value × hours worked.

33 **Four considerations for selecting a base (SM 4.17)?**

A Economy, practicability, equitability and reliability.

5.4 Absorption rates and the machine hour rate

34 **Name ICAI's six methods of absorbing overheads, in order.**

A Percentage of direct materials, percentage of prime cost, percentage of direct labour cost, labour hour rate, machine hour rate, rate per unit of output.

35 **Why does ICAI call the percentage of material cost and of prime cost generally unsuitable?**

A Overheads are mostly a function of time; they are hardly related to material or prime cost; material prices fluctuate widely; skill and use of machines are ignored.

36 **Direct versus comprehensive machine hour rate?**

A Direct: only expenses directly connected with operating the machine, divided by its operational hours. Comprehensive: also a share of the department's other manufacturing expenses (supervision, lighting, rent, consumables), sometimes the operator's wages.

37 **Standing charges versus machine expenses: give two of each.**

A Standing charges (fixed): insurance, rent, supervisor's salary, depreciation on time basis. Machine expenses (variable): power, consumables, repairs and maintenance, depreciation on activity basis.

38 **Three advantages of the machine hour rate?**

A Best where machines are the main factor of production; under-absorption shows the extent of machine idleness; advantageous where one operator attends several machines or several operators work one machine.

39 **Conditions for applying a blanket overhead rate?**

A Only one major product; or several products where all products pass through all departments and are processed for the same length of time in each department.

40 **Formula: pre-determined overhead rate?**

A Budgeted amount of overheads ÷ budgeted base.

41 **Three bases on which a pre-determined rate may be estimated?**

A The previous year's figure; estimated expenses and anticipated volume of production; the normal volume of business.

42 **ICAI's definition of normal capacity?**

A The volume of production or services achieved or achievable on an average over a period under normal circumstances, taking into account the reduction in capacity resulting from planned maintenance.

43 **Normal idle capacity and abnormal idle capacity?**

A Normal idle = installed capacity − normal capacity. Abnormal idle = normal capacity − actual capacity, where actual is lower.

44 **On what capacity are fixed manufacturing overheads absorbed, and variable ones?**

A Fixed on normal capacity; variable on actual production.

45 **Operator-limited machine hours: wages on which hours, rate on which hours?**

A Wages on paid hours (available less absenteeism without pay); the rate on effective hours (paid hours less leave with pay and normal idle time).

5.5 Under- and over-absorption of overheads

46 **Define under-absorption and over-absorption.**

A Under-absorption: the absorbed amount is less than the actual amount (debit balance). Over-absorption: the absorbed amount is more than the actual amount (credit balance). Either may be termed overhead variance.

47 **Formula: supplementary rate?**

A Under- or over-absorbed overheads ÷ units produced (finished units plus equivalent completed units in WIP).

48 **Which accounts receive the supplementary rate, and on what quantities?**

A Stock of finished goods (units in stock), stock of semi-finished goods or WIP (equivalent completed units) and cost of sales (units sold); debited for under-absorption, credited for over-absorption.

49 **Treatment of a small under- or over-absorbed balance?**

A Transferred to the Costing Profit and Loss Account; product costs are not adjusted, as they represent normal cost.

50 **Treatment of a large under-absorption due to wrong estimation for unavoidable reasons?**

A Adjust product costs through a supplementary rate over cost of sales, finished goods and work-in-progress.

51 **Treatment of under- or over-absorption due to abnormal reasons?**

A Transferred to the Costing Profit and Loss Account (for example abnormal over- or under-capacity utilisation, defective planning, factory inefficiency).

52 **Name four items ICAI's answers deduct from actual overheads before computing under-absorption.**

A Wages paid for a strike period under an award; amounts paid under a labour court award (non-recurring); expenses of the previous year booked in the current year; obsolete stores written off.

53 **Treatment of idle capacity costs: three cases?**

A Unavoidable reasons (repairs, maintenance, changeover): supplementary overhead rate on capacity utilised. Avoidable reasons (faulty planning, power failure): Costing P&L. Seasonal factors: charged to cost of production by inflating overhead rates.

54 **When do actual and pre-determined rates give neither under- nor over-absorption?**

A When actual overheads and actual units change in the same proportion (or both equal the budget).

55 **Journal entry for under-absorption spread by a supplementary rate, with an abnormal part?**

A WIP Control A/c, Finished Goods Control A/c, Cost of Sales A/c and Costing P&L A/c Dr.; To Overhead Control A/c.

5.6 Administration, selling, distribution and R&D overheads

56 **CIMA's definition of administrative overhead?**

A The sum of those costs of general management and of secretarial accounting and administrative services, which cannot be directly related to the production, marketing, research or development functions of the enterprise.

57 **Three methods of accounting for administrative overheads, in ICAI's order?**

A (a) Apportioning between production and sales departments; (b) charging to Costing Profit and Loss Account; (c) treating them as a separate addition to cost of production or sales.

58 **Bases for absorbing administrative overheads as a separate addition to cost?**

A Works cost; sales value or quantity; gross profit on sales; quantity produced; conversion cost, etc.

59 **Two disadvantages of charging administrative overheads to Costing P&L?**

A Cost of products is understated; excluding them from product cost is against sound accounting principle.

60 **Three methods of controlling administrative overheads?**

A Classification and analysis by departments according to functions compared with results accomplished; control through budgets; control through standards.

61 **Four bases for absorbing selling and distribution overheads, and the most logical?**

A Sales value of goods, cost of goods sold, gross profit on sales, number of orders or units sold; sales value is ordinarily the most logical.

62 **Three reasons controlling selling and distribution overheads is difficult?**

A Incidence depends on external factors beyond management's control; dependence on customers' behaviour and tastes; they are policy costs not amenable to control.

63 **Three methods of controlling selling and distribution overheads?**

A Comparison with past performance; budgetary control; standard costing.

64 **Treatment of development costs of a particular product?**

A Charged directly to that product, usually as deferred revenue expenses, recovered as a cost per unit when production is fully established.

65 **Treatment of research that proves a failure?**

A Excluded from costs and charged to the Costing Profit and Loss Account.

66 **Treatment of interest and financing charges in a cost statement?**

A Presented as a separate item of cost of sales; imputed costs are not included.

67 **Treatment of carriage and cartage?**

A On direct material: cost of direct material; on indirect material (stores): factory overheads; on finished goods: distribution overhead.

CHAPTER 6

Activity Based Costing

6.1 Activities, cost pools, cost drivers and the cost hierarchy

1 **ABC: define a cost driver and name its two categories.**

A A factor that causes a change in the cost of an activity. Resource cost driver (quantity of resources consumed by an activity) and activity cost driver (frequency and intensity of demand placed on activities by cost objects).

2 **ABC: what is a resource cost driver used for, and what is an activity cost driver used for?**

A A resource cost driver assigns the cost of a resource to an activity or cost pool; an activity cost driver assigns activity costs to cost objects.

3 **ABC: what is a cost pool?**

A A group of individual cost items that have the same cause and effect relationship, e.g. machine set-up.

4 **Formula: activity cost driver rate?**

A Total cost of activity ÷ activity driver (total quantity of the driver for the period).

5 **The four levels of the cost hierarchy, in ICAI's order?**

A Unit level, batch level, product level, facility level activities.

6 **Batch-level activities: how does their cost behave?**

A It varies with the number of batches made, but is common (fixed) for all units within the batch, e.g. machine set-up, material ordering per batch.

7 **Two examples each of product-level and facility-level activities (SM)?**

A Product level: designing the product, producing parts specifications, keeping technical drawings up to date. Facility level: maintenance of buildings, plant security.

8 **The four situations in which ABC is particularly useful (SM 5.3)?**

A High amount of overhead; wide range of products; presence of non-volume related activities; stiff competition.

9 **The four factors that prompted the development of ABC?**

A Growing overheads from automation; increasing competition; increasing product diversity; decreasing costs of information processing.

10 **In which stage do ABC and traditional absorption costing differ?**

A The first stage: ABC relates overheads to activities (cost pools) instead of functional departments (cost centres); both then charge cost objects.

11 **Cost drivers for distribution and for customer service (SM)?**

A Distribution: number of units distributed, number of customers. Customer service: number of service calls, number of products serviced, hours spent on servicing products.

6.2 Product costs under ABC versus traditional absorption

12 **ABC: formula for cost distortion per unit, and what its sign means?**

A Traditional cost per unit – ABC cost per unit. Positive: over-costed by the traditional system; negative: under-costed.

13 **ABC: what is the total of the cost distortions across all products?**

A Nil, when both systems absorb the same total overheads: over-costing of some products equals under-costing of the others.

14 **Which cost elements are unchanged when a firm moves from traditional absorption to ABC?**

A Prime cost: direct materials, direct labour and direct expenses. Only production overheads are re-spread.

15 **ABC: how are batches and per-batch drivers turned into totals?**

A Batches = units produced ÷ batch size; total orders (or inspections) = batches × orders (or inspections) per batch.

16 **Formula: cost of unused capacity under capacity-based activity rates?**

A Cost of the activity − cost allocated to products, or unused driver units × rate (rate = cost ÷ driver capacity).

17 **The six factors in choosing a capacity level for the budgeted fixed overhead rate (SM 5.23)?**

A Effect on product costing and capacity management; on pricing decisions; on performance evaluation; on financial statements; regulatory requirements; difficulties in forecasting.

18 **Format: customer-level operating income?**

A Revenue at list price – discount = revenue at actual price; – cost of goods sold = gross margin; – customer-level operating activity costs = customer-level operating income.

19 **Customer profitability: operating income as a % of which revenue?**

A Revenue at actual price (net of discount), as in ICAI's suggested answers.

20 **Five challenges in assigning operating costs to activity areas (SM 5.49)?**

A Choosing an appropriate cost driver; developing a reliable database for it; handling costs common to several activities; choice of time period for cost rates; behavioural factors.

21 **ICAI's conclusion in Illustration 3 about volume and cost distortion?**

A The low volume product is under-costed and the high volume product over-costed when direct labour hours are used to absorb overheads.

6.3 Activity based management and budgeting; benefits and limits

22 **ABM: ICAI's meaning?**

A The cost management application of ABC: using ABC as a costing tool to manage costs at activity level; a discipline that focuses on the efficient and effective management of activities as the route to continuously improving the value received by customers; it utilizes cost information gathered through ABC.

23 **The three types of analysis involved in ABM?**

A Cost driver analysis; activity analysis (value-added and non-value-added activities); performance analysis.

24 **The four ways ABM can be used in business?**

A Cost reduction; business process re-engineering; benchmarking; performance measurement.

25 **Value-added and non-value-added activities: ICAI's examples?**

A Value-added: polishing furniture by a furniture manufacturer. Non-value-added: moving materials, and machine set-up for a production run.

26 **Benchmarking under ABM: what is it and what does it require?**

A Comparing ABC-derived activity costs of one segment of a company with those of other segments; it requires uniformity in the definition of activities and the measurement of their costs.

27 **The four limitations of ABC (SM 5.18)?**

A More expensive than traditional costing; not helpful to small organisations; may not be applied to organisations with limited products; selecting the most suitable cost driver may be difficult.

28 **The five practical stages of ABC implementation?**

A Staff training; process specification; activity definition; activity driver selection; assigning cost.

29 **ABB: how does ICAI describe its relation to ABC?**

A ABB is the reversing of the ABC process to produce financial plans and budgets.

30 **The three key elements of activity based budgeting?**

A Type of work to be done; quantity of work to be done; cost of work to be done.

31 **Three benefits of activity based budgeting?**

A Enhances the accuracy of financial forecasts and management understanding; when automated, rapidly produces plans for varying volume assumptions; eliminates much of the needless rework of traditional budgeting.

32 **Name four of ICAI's seven advantages of ABC.**

A Any four of: more accurate costing; logical overhead allocation; better pricing policies; utilizes unit cost rather than total cost; identifies non-value added activities; helpful to multi-product organisations; highlights problem areas.

CHAPTER 7

Cost Accounting Systems

7.1 Non-integrated and integrated accounting systems

1 **Under the non-integrated system, which account completes the double entry, and what else is it called?**

A The Cost Ledger Control Account, also known as the General Ledger Adjustment Account.

2 **Cost Ledger Control A/c: what is debited and credited to it, and what does its balance represent?**

A Credited with all items of expenditure; debited with sales; the costing profit or loss is transferred to it. Its balance is the net total of all the balances of the impersonal accounts.

3 **The four important ledgers under the non-integrated system?**

A Cost ledger, stores ledger, work-in-process (job) ledger, finished goods ledger.

4 **Non-integrated system: journal entry for materials returned to the vendor?**

A Cost Ledger Control A/c Dr. / To Stores Ledger Control A/c.

5 **Non-integrated system: journal entry for materials purchased for a special job?**

A Work-in-Process (Job) Ledger Control A/c Dr. / To Cost Ledger Control A/c.

6 **Journal entry for over-absorbed production overhead transferred to Costing P&L?**

A Production Overhead Control A/c Dr. / To Costing Profit & Loss A/c.

7 **Four advantages of integrated accounts?**

A No need for reconciliation; less efforts; less time consuming; economical process (centralisation of accounting function).

8 **Four essential pre-requisites for integrated accounts?**

A Management's decision on the extent of integration; a suitable coding system; an agreed routine for accruals, prepaid expenses and other adjustments; perfect coordination between cost and financial staff.

9 **Which accounts replace the Cost Ledger Control A/c in integrated accounts?**

A Bank, receivables (debtors), payables (creditors), provision for depreciation, fixed assets and share capital accounts.

10 **Integrated accounts: entry for credit sales?**

A Receivables (Debtors) A/c Dr. / To Sales A/c.

11 **Incomplete control accounts: how are materials purchased found from the creditors' data?**

A Payments to creditors + closing creditors – opening creditors.

7.2 Reconciling cost and financial profit

12 **Reconciliation of cost and financial accounts: the three steps (SM 7.33)?**

A Ascertainment of profit as per financial accounts; ascertainment of profit as per cost accounts; reconciliation of both the profits.

13 **When can the reconciliation statement be avoided?**

A When the cost and financial accounts are integrated: the same set of accounts fulfils the requirements of both.

14 **The four causes of differences between financial and cost accounts, in ICAI's order?**

A Items in financial accounts only; items in cost accounts only (notional); items treated differently in the two sets; varying basis of valuation.

15 **Purely financial expenses: ICAI's six?**

A Interest on loans or bank mortgages; expenses and discounts on issue of shares, debentures etc.; other capital losses (e.g. uninsured fire loss); losses on sale of fixed assets and investments; income tax, donations, subscriptions; expenses of the share transfer office.

16 **Purely financial income: ICAI's five?**

A Interest received on bank deposits, loans and investments; dividends received; profits on sale of fixed assets and investments; transfer fee received; rent receivables.

17 **Items in cost accounts only (notional): ICAI's four?**

A Charges in lieu of rent where premises are owned; notional interest on capital; notional salary of the proprietor; notional depreciation on fully depreciated assets.

18 **Stock valuation as a cause of difference?**

A Financial accounts value stock at cost or market price, whichever is lower; cost accounts value it only at cost.

19 **From profit as per cost accounts: is under-absorbed overhead added or deducted?**

A Deducted: actual overheads exceeded the amount charged in the cost accounts, so the financial profit is lower.

20 **From profit as per cost accounts: is closing stock over-valued in cost accounts added or deducted? And opening stock over-valued?**

A Closing stock over-valued in cost accounts: deducted. Opening stock over-valued in cost accounts: added.

21 **Memorandum Reconciliation Account with a costing profit as base: where do the base and the balancing figure go?**

A Profit as per cost accounts on the credit side with the additions; deductions on the debit side; profit as per financial accounts as the balancing figure on the debit side.

CHAPTER 8

Unit and Batch Costing

8.1 Unit costing

1 **ICAI's definition of unit costing?**

A The method of costing where the output produced is identical and each unit of output requires identical cost; also known as single or output costing.

2 **Formula for cost per unit in unit costing?**

A Total cost of production ÷ number of units produced.

3 **Five industries ICAI names for unit costing?**

A Paper, cement, steel works, mining, breweries.

4 **Methods of costing for job-work industries and for continuous or process industries (SM 8.2)?**

A Job work: job, contract and batch costing. Continuous or process type: process costing, single output costing, operating costing.

5 **Unit costing: treatment of normal defectives within the established rate?**

A The cost of rectification or loss is charged to the entire output.

6 **Unit costing: treatment of defectives substantially exceeding normal limits, and of losses due to abnormal reasons?**

A The cost beyond normal limits, and the whole loss from abnormal reasons, is written off in the Costing Profit and Loss Account.

7 **Source documents for collecting material and labour cost in unit costing?**

A Material requisition notes for materials; job time cards or sheets and idle time cards for direct labour; payroll books for indirect labour.

8 **How do you split a semi-variable cost from two output levels?**

A Variable rate = change in cost ÷ change in output; fixed part = total cost − output × variable rate.

9 **Average selling price for a production programme?**

A (Total cost of all periods + desired profit) ÷ total units of all periods.

10 **"Extra ₹50,000 for every 15% increase in capacity or part thereof": how many steps from 40% to 75%?**

A Three (two full 15% steps and part of a third).

8.2 Batch costing and the economic batch quantity

11 **ICAI's definition of batch costing?**

A A type of specific order costing where articles are manufactured in predetermined lots, known as batch; the cost object is a batch for production rather than output as in unit costing.

12 **Costing procedure in batch costing (SM 8.8)?**

A One number for each batch; material cost from material requisitions for the batch; labour cost = time on the batch (time cards or job tickets) × rate; overheads absorbed on machine hours, direct labour hours etc.

13 **Define economic batch quantity.**

A The size of a batch where the total cost of set-up and holding costs are at a minimum.

14 **EBQ formula and its symbols?**

A $\sqrt{(2DS \div C)}$: D = annual demand for the product, S = setting-up cost per batch, C = carrying cost per unit of production per annum.

15 **The two main costs whose total EBQ minimises?**

A Machine set-up costs and inventory holding costs.

16 **Two ways of determining EBQ?**

A Calculating the total cost for a series of possible batch sizes and choosing the minimum, or the formula $\sqrt{(2DS \div C)}$.

17 **At the EBQ, how do annual set-up cost and annual holding cost compare?**

A They are equal.

18 **Interval between two runs?**

A 365 days ÷ number of runs a year (ICAI rounds the runs up).

19 **Job costing vs batch costing: the three points of SM 8.15?**

A (1) Non-standard, non-repetitive products to customer specification vs homogeneous products in lots; (2) cost determined for each job vs in aggregate for the batch, then per unit; (3) jobs unique vs batch products lack individuality.

20 **Holding cost ₹1 per unit per month: what C goes into the EBQ formula?**

A ₹12 per unit per annum.

CHAPTER 9

Job Costing

9.1 Job costing

1 CIMA's definition of job costing?

A The category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches, each of which is authorised by specific order or contract.

2 When is job costing suitable (SM 9.3)?

A When jobs are executed for different customers to their specifications; when no two orders are alike and each needs special treatment; where work-in-progress differs from period to period with the number of jobs in hand.

3 Name five industries ICAI gives for job costing.

A Printing, furniture, hardware, ship-building, heavy machinery, interior decoration, repairs (any five).

4 Document for surplus material returned from a job to stores, and for material moved to another job?

A Returned: stores debit note or shop credit (the job is credited). Moved: material transfer note naming both jobs.

5 **How is special material bought for a particular job treated?**

A Charged direct to that job as soon as it is purchased, without passing through the Stores Ledger.

6 **Entry for overhead charged to jobs?**

A Work-in-Process Control A/c Dr. To Factory Overhead Control A/c.

7 **Entry for the sales value of jobs completed?**

A Cost Ledger Control A/c Dr. To Costing Profit and Loss A/c.

8 **Another name of the Cost Ledger Control Account?**

A General Ledger Adjustment Account.

9 **ICAI's five advantages of job costing?**

A Cost details of each job available for control; profitability of each job; facilitates production planning; budgetary control and standard costing can be applied; spoilage and defectives identified and responsibility fixed.

10 **ICAI's four disadvantages of job costing?**

A Costly and laborious; much clerical work so more chance of error; not suitable in inflationary conditions; previous cost records meaningless if market conditions change.

11 **Treatment of overtime premium paid because the customer wanted the job early?**

A Charged to that job directly.

CHAPTER 10

Process and Operation Costing

10.1 Process accounts, normal loss, abnormal loss and abnormal gain

1 **ICAI's definition of process costing?**

A A method of cost accounting whereby costs are charged to processes or operations and averaged over units produced.

2 **Formula: cost per unit of good output in a process with normal loss?**

A (Total cost − realisable value of normal loss) ÷ (total input units − normal loss units).

3 **Treatment of normal process loss in cost accounts?**

A Its cost is absorbed by the good units produced; the amount realised by selling the normal loss units is credited to the process account.

4 **Treatment of abnormal process loss in cost accounts?**

A Each unit is valued at the cost of a good unit; the total is credited to the process account and debited to the Costing Profit and Loss Account (via the abnormal loss account, net of scrap realised).

5 **Treatment of abnormal gain in cost accounts?**

A The process account is debited and the abnormal gain account credited at the normal cost per unit; the abnormal gain account is closed to the Costing P&L after deducting the scrap value the gain units would have fetched.

6 **How is the value of normal loss computed?**

A Normal loss units × (scrap realisable value less cost to sell); it is calculated on the estimated (normal) figure even when the actual loss is less.

7 **ICAI: three causes of abnormal process loss?**

A Carelessness of workers, a bad plant design or operation, sabotage (it cannot be estimated in advance but can be controlled).

8 **ICAI: three causes of abnormal gain?**

A Over-estimation of process loss, improvement in work efficiency of workers, use of better technology in production.

9 **Are units withdrawn for test or sampling a normal or an abnormal loss?**

A Normal process loss.

10 **What does the Abnormal Gain Account debit before the balance goes to the Costing P&L?**

A The normal loss account, with the scrap value of the abnormal gain units (gain units × scrap rate).

11 **Process costing: six basic features, in ICAI's order?**

A Plant divided into processes; continuous manufacture; output of one process is input of another; like units; no tracing of output lots to input lots; may give rise to joint and by-products.

10.2 Equivalent units: FIFO and weighted average

12 **ICAI's formula for equivalent completed units?**

A Actual number of units in the process of manufacture × percentage of work completed.

13 **FIFO: equivalent units of opening WIP?**

A Opening WIP units × the percentage of work still to be done in the current period (100% minus the percentage done earlier).

14 **Under which method is closing WIP valued at current cost?**

A FIFO.

15 **Weighted average: what is divided by the equivalent units?**

A The cost of opening work-in-process plus the cost of the current period (less the scrap value of normal loss).

16 **Equivalent units for normal loss?**

A None; normal loss is given no equivalent units.

17 **How is abnormal gain treated in the statement of equivalent production?**

A As 100% complete for all cost elements, irrespective of the percentage of completion, and deducted.

18 **Finished output in the statement of equivalent units: FIFO vs weighted average?**

A FIFO: units completed and transferred less opening WIP. Weighted average: all units completed and transferred.

19 **Main difference between FIFO and weighted average (ICAI)?**

A Under average, units of opening WIP and their cost are taken in full; under FIFO only the remaining work done now is considered.

20 **The five steps of a production cost report, in order?**

A Physical flow of units; equivalent units for each element; total cost for each element; cost per equivalent unit; assignment of cost to completed units, WIP and abnormal loss.

21 **Link between weighted average and FIFO equivalent units?**

A EU (weighted average) − EU (FIFO) = opening WIP units × percentage completed in the previous period, for each element.

10.3 Inter-process profit and operation costing

22 ICAI: what are inter-process profits?

A The difference between cost and the transfer price when the output of one process is transferred to the next not at cost but at market value or cost plus a percentage of profit.

23 Inter-process profit: ICAI's two advantages?

A Comparison between the cost of output and its market price at the stage of completion is facilitated; each process is made to stand by itself as to the profitability.

24 Inter-process profit: ICAI's two disadvantages?

A It involves complication; the system shows profits which are not realised because of stock not sold out.

25 Transfer price when profit is p% of the transfer price?

A Cost transferred ÷ (1 – p); e.g. cost ₹75 at 25% on TP gives TP ₹100.

26 Profit of 1/3 on cost is what fraction of the transfer price?

A 1/4 (cost 3 + profit 1 = TP 4).

27 **Stock in process valued at prime cost: how is its cost element found?**

A Closing stock × prime cost (cost column) ÷ prime cost (total column, including inter-process profit).

28 **How is the profit actually realised found in inter-process profit accounts?**

A Sales – actual cost of goods sold; equivalently, profits shown + profit in opening stocks – profit in closing stocks.

29 **Why is operation costing called a hybrid product costing system?**

A Material costs are accumulated by job order or batch, while conversion costs are accumulated by department and assigned by process costing methods.

30 **Operation costing: when is it used, and three industries (ICAI)?**

A When more than one variant of a product uses different materials but similar conversion activities; ready-made garments, shoe making, jewellery.

31 **Operation costing: on what basis are conversion costs applied to products?**

A A predetermined application rate based on budgeted conversion costs.

CHAPTER 11

Joint Products and By-Products

11.1 Apportioning joint costs: physical units, sales value and net realisable value

1 **ICAI's definition of joint products?**

A Two or more products separated in the course of the same processing operation usually requiring further processing, each in such proportion that no single product can be designated as a major product.

2 **What is the split-off point, and what happens to costs after it?**

A The point in a production process where joint products become separately identifiable. Joint cost is incurred up to it; any cost after it is product-specific and borne by the product concerned.

3 **Three things affected by the apportionment of joint cost (ICAI)?**

A (a) Valuation of closing inventory; (b) pricing of products; (c) profit or loss on the sale of different products.

4 **ICAI's three commonly used methods of apportioning joint cost?**

A (i) Physical units method; (ii) net realisable value at split-off point; (iii) using technical estimates.

5 **ICAI's four "other methods" managers may use for decisions?**

A (i) Market value at the point of separation; (ii) market value after further processing; (iii) average unit cost method; (iv) contribution margin method.

6 **Formula: multiplying factor (market value at the point of separation)?**

A Joint cost ÷ total sales revenue at split-off × 100; each product's joint cost = its sales value × the factor.

7 **What is deducted from final sales value to get NRV at split-off?**

A Profit margin (if given), directly attributable selling and distribution costs (freight, royalty, commission), and post split-off processing cost.

8 **Contribution margin method: what is apportioned on what basis?**

A Variable joint cost on physical units; fixed joint cost in the ratio of contributions (sales – variable cost, including variable cost after split-off).

9 **Main defect of the physical units method (ICAI)?**

A It gives equal importance and value to all the joint products; it also cannot be used if the products cannot be converted to a common unit.

10 **When is the market value at the point of separation method used (ICAI's circumstances)?**

A Where further processing costs are incurred disproportionately.

11 **Co-products: definition and example?**

A Two or more products which are contemporary but do not necessarily emerge from the same material in the same process, e.g. wheat and gram grown on two separate farms.

11.2 Sell at split-off or process further

12 Sell at split-off or process further: what is the decision rule?

A Process further only if incremental revenue (sales after further processing – sales value at split-off) exceeds the incremental cost of further processing.

13 Why is the joint cost irrelevant to the decision to process a joint product further?

A It is incurred before the split-off point and is the same whether the product is sold at split-off or processed further; a relevant cost must be a future cost that differs between the options.

14 Formula: incremental revenue from further processing?

A Units after further processing (after any loss) × final price – units at split-off × split-off price.

15 What goes into the incremental cost of further processing?

A Further processing materials, labour and variable overheads; fixed costs that arise only because of the processing; additional selling, packing or marketing costs; less selling expenses saved.

16 **How does a loss in further processing enter the decision?**

A It reduces the quantity sold after processing: final sales = (units − loss) × final price, compared with all the units at the split-off price.

17 **ICAI's wording of the advice in a further-processing question?**

A "It is advisable to further process only product ___ and to sell product ___ at the point of separation."

18 **Formula: minimum final price that justifies further processing?**

A (Sales value at split-off + incremental cost) ÷ output after further processing.

19 **A buyer offers a price for a joint product at the split-off point. How is the offer evaluated?**

A Compare the revenue after further processing with the revenue from the offer, and the further processing and selling costs with the costs of the offer; accept if processing adds less than it costs.

20 **When does the joint cost become relevant?**

A When deciding whether to run the joint process at all: the revenue of the best plan must cover the joint cost plus the further costs of that plan.

11.3 Treatment of by-products

21 **ICAI's definition of by-products?**

A Products recovered from material discarded in a main process, or from the production of some major products, where the material value is to be considered at the time of severance from the main product.

22 **Two points of distinction between joint products and by-products (ICAI)?**

A (a) Joint products are of equal importance whereas by-products are of small economic value; (b) joint products are produced simultaneously but by-products are produced incidentally in addition to the main products.

23 **ICAI's examples of by-products?**

A Molasses in the manufacture of sugar; tar, ammonia and benzole on carbonisation of coal; glycerin in the manufacture of soap.

24 **ICAI's four methods of apportioning joint cost to by-products, in order?**

A (i) Net realisable value method; (ii) standard cost in technical estimates; (iii) comparative price; (iv) re-use basis.

25 **By-product of small total value: the two treatments?**

A (1) Credit its sales value to the Costing Profit and Loss Account (as miscellaneous income or additional sales revenue), with no credit in the cost accounts; (2) deduct its sale proceeds from the total costs, from production cost or cost of sales.

26 **By-product of considerable total value: treatment?**

A Regard it as a joint product; apportion the costs up to the point of separation over the by-products and joint products by relative market values, physical output at split-off, or ultimate selling prices.

27 **By-product that requires further processing: treatment?**

A Find its NRV at split-off (realisable value – further processing cost); if the total is small treat it as a small-value by-product, if considerable as a joint product.

28 **When is the standard cost in technical estimates method used for a by-product?**

A Where the by-product is not saleable in the condition in which it emerges, or comparative prices of similar products are not available.

29 **Comparative price method for a by-product: what is it, with ICAI's example?**

A The by-product is valued at the price of a similar or alternative material: gas from a blast furnace used in the factory is valued at what a gas company would have charged.

30 **Where does the by-product credit appear in the cost sheet?**

A "Less: Credit for Recoveries/Scrap/By-Products/misc. income", deducted before arriving at the cost of production; the NRV of by-products is deducted from the cost of production.

CHAPTER 12

Service Costing

12.1 Service costing and composite cost units

1 **What is a composite cost unit? Give four examples.**

A Two measurement units combined to know the cost of a service or operation: tonne-km, quintal-km, passenger-km, patient-day.

2 **In how many ways may a composite unit be computed? Name them.**

A Two: the absolute (weighted average) basis and the commercial (simple average) basis.

3 **Formula: absolute tonne-km?**

A $(W_1 \times D_1) + (W_2 \times D_2) + \ldots + (W_n \times D_n)$: weight carried × distance for each leg, added up.

4 **Formula: commercial tonne-km?**

A $(D_1 + D_2 + \ldots + D_n) \times (W_1 + W_2 + \ldots + W_n) \div n$: total distance × the simple average load.

5 **In goods transport, which factor is quantitative and which qualitative?**

A Distance is quantitative (the variable cost depends on it); weight carried is qualitative.

6 **What is an equivalent service unit?**

A Different grades of a service using common resources are each given a weight and converted into equivalent units, making them equivalent and comparable.

7 **Cost units: hospital, hotel, electricity, insurance, IT?**

A Hospital: patient per day, room per day, per bed, per operation. Hotel: guest days or room days. Electricity: kWh. Insurance: per policy, per claim, per TPA. IT and ITES: per project, per module.

8 **What is a Key Performance Indicator?**

A The quantitative and qualitative factors commonly used to assess the performance of an organisation, which are important to achieve its goal.

9 **Three hotel KPIs in the Study Material?**

A Cost per occupied room (CPOR), occupancy rate, revenue per available room (RevPAR).

10 **Service costing is also known as?**

A Operating costing.

12.2 Transport costing: passenger-km and tonne-km

11 **Cost unit of a goods transport and of a passenger transport organisation?**

A Goods: tonne-km (cost of carrying one tonne one kilometre). Passengers: passenger-km (cost of carrying one passenger one kilometre).

12 **Formula: passenger-km of a bus service?**

A Number of buses × distance × seating capacity × used capacity × number of days × number of single trips.

13 **ICAI's standing charges of a transport undertaking?**

A Insurance; licence fees; salaries of driver, conductor, cleaners if paid monthly; garage costs including rent; depreciation related to efflux of time; taxes; administration expenses.

14 **ICAI's running costs of a transport undertaking?**

A Petrol and diesel; lubricant oils; wages of driver, conductor, cleaners if related to operations; depreciation related to activity; any other variable costs.

15 **ICAI's maintenance (semi-variable) costs of a transport undertaking?**

A Repairs and maintenance; tyres; spares.

16 **Is a driver's salary a standing or a running charge?**

A Either, depending on the situation and the nature of his employment: standing if paid monthly, running if related to operations.

17 **Takings when commission is 10% and profit 15% of takings?**

A Takings = cost before commission ÷ (1 − 0.10 − 0.15) = cost ÷ 0.75.

18 **Treatment of a penalty paid for overloading a lorry?**

A Abnormal: excluded from operating cost, debited to Costing Profit and Loss Account and deducted from operating profit. The actual (overloaded) weight still counts for tonne-km.

19 **Fare per passenger for a journey?**

A Rate per passenger-km (takings ÷ passenger-km) × the journey's kilometres.

20 **Diesel cost of a vehicle for a period?**

A Km run ÷ km per litre × price per litre.

12.3 Hotels, hospitals, power, IT and other services

21 **Cost unit of a hotel, and what is taken into account in computing it?**

A Guest-day or room-day, with the estimated occupancy rate at different points of time (peak season, lean season).

22 **Units of cost for a hospital's out-patient, in-patient, scanning and laundry departments?**

A Per out-patient; per room-day; per case; per 100 items laundered.

23 **Hospital cost segregation: examples of fixed and variable costs?**

A Fixed (based on timelines): staff salaries, depreciation on building and equipment. Variable: laundry, food supplied to patients, power.

24 **What is BOT?**

A Build-Operate-Transfer: the private sector designs, finances, constructs and operates the facility and, after the specified concession period, transfers ownership to the Government.

25 **Capital costs of a toll road project?**

A Preliminary and pre-operative expenses, land acquisition, materials, labour, overheads in actual construction, contingency allowance, interest during construction period.

26 **Operating and maintenance costs of a toll road?**

A Toll collection expenses; administrative expenses for day-to-day operation; maintenance (routine and periodic); interest for servicing term loans.

27 **Toll rate formula in the Study Material?**

A User fee = total distance × toll rate per km, rounded off to the nearest multiple of five.

28 **IT and ITES: what is effort estimation, and how is effort costed?**

A Identifying the skill level and duration of each activity of a project; effort is costed per person-day, person-week or person-month.

29 **IT and ITES: when are hardware, software, travel and training a direct project cost?**

A When identifiable with (incurred for) the project; otherwise they are service overhead.

30 **Insurance ABC: the two groups of activities?**

A Pre-product development (market research, product development) and post-product development (selling of policy; processing of claims).

31 **Cost unit of a power house generating electricity?**

A Kilowatt-hour (kWh), commonly known as a "Unit".

32 **Four kinds of cost centre in an educational institution?**

A Primary or direct (departments); service (laboratory, library, sports); students' self-supporting services (transport, hostel and mess); administration (research and improvement, examination).

CHAPTER 13

Standard Costing

13.1 Standards, their types and standard costing

1 **CIMA definition of standard costing (SM)?**

A "Control technique that reports variances by comparing actual costs to pre-set standards so facilitating action through management by exception."

2 **CIMA definition of standard cost (SM)?**

A "The planned unit cost of the product, component or service produced in a period. The standard cost may be determined on a number of bases. The main use of standard costs is in performance measurement, control, stock valuation and in the establishment of selling prices."

3 **The five steps in the process of standard costing?**

A Setting of standards; ascertainment of actual costs; comparison of actual cost with standard cost; investigating the reasons for variances; disposition of variances.

4 **The four types of standards, in ICAI's order?**

A Ideal, normal, basic (bogey) and current standards.

5 **Three criticisms of ideal standards?**

A Unattainable, so no one takes them seriously; the variances do not show how far costs could reasonably and practically have been avoided; there is no logical method of disposing of the variances.

6 **How are basic standards used, since variances are not calculated?**

A Actual cost and current standard cost are expressed as percentages of basic cost and compared with each other and with previous periods to show the trend.

7 **Which type of standard is most effective for cost control (SM)?**

A One which represents an attainable level of good performance.

8 **Four steps in setting material quantity standards?**

A Standardisation of products; product study; preparation of specification list; test runs.

9 **Six steps in setting labour time standards?**

A Standardisation of product and product study; labour specification; standardisation of methods; manufacturing layout; time and motion study; training and trial.

10 **Four factors for material price standards when price fluctuations are small?**

A Stock on hand and its prices; prices of orders already placed for future deliveries; minimum support price fixed by the appropriate authority; anticipated fluctuation in price levels.

11 **Controllable vs uncontrollable variance (SM)?**

A Controllable: can be controlled under normal operating conditions if the responsibility centre takes preventive measures and acts prudently. Uncontrollable: due to conditions beyond its control, not controllable even with all preventive measures.

12 **What is a revision variance?**

A The difference between the original standard cost and the revised standard cost of actual production.

13.2 Material variances

13 **Formula: material cost variance?**

A (Standard quantity for actual output × standard price) − (actual quantity × actual price): the standard cost of actual output minus the actual cost.

14 **Formula: material price variance, and which actual quantity it uses?**

A AQ × (SP − AP), where AQ is the quantity purchased; if purchases are not given, it is taken as equal to the quantity consumed.

15 **Formula: material usage variance, and which actual quantity it uses?**

A SP × (SQ − AQ), where AQ is the quantity actually used.

16 **Formula: material mix variance?**

A SP × (RSQ − AQ): actual quantity in standard proportion minus actual quantity in actual proportion, at standard price.

17 **Formula: material yield (sub-usage) variance?**

A SP × (SQ − RSQ): standard quantity for actual production minus actual quantity in standard proportion, at standard price.

18 **What is the revised standard quantity (RSQ)?**

A The actual total quantity of inputs used, split in the standard proportion.

19 **When does a material mix variance arise?**

A Only when two or more inputs are used to produce a product.

20 **Material variances: the two verification formulas?**

A MCV = MPV + MUV (if quantity purchased equals quantity consumed); MUV = MMV + MYV.

21 **Standard input for 1,700 kg of output with a 15% standard loss?**

A 1,700 × 100 ÷ 85 = 2,000 kg, split by the standard mix.

22 **Who is responsible for material price and material usage variances?**

A Price: the purchase department (for adverse variances from factors it controls). Usage: the production department.

23 **Three causes of a material yield variance (SM)?**

A Sub-standard quality of materials, inefficiency of workers, wrong processing.

13.3 Labour variances, including idle time

24 **Formula: labour cost variance?**

A (Standard hours for actual output × standard rate) − (actual hours paid × actual rate).

25 **Formula: labour rate variance, and which hours does it use?**

A Actual hours paid × (standard rate − actual rate).

26 **Formula: labour efficiency variance, and which hours does it use?**

A Standard rate × (standard hours for actual production − actual hours worked).

27 **Formula: labour idle time variance?**

A Standard rate × (actual hours paid − actual hours worked) = standard rate × idle hours.

28 **Formula: labour mix (gang) variance?**

A Standard rate × (revised standard hours − actual hours worked).

29 **Formula: labour yield (sub-efficiency) variance?**

A Standard rate × (standard hours – revised standard hours).

30 **What are revised standard hours (RSH)?**

A The actual hours worked, if worked in the standard mix (combination) of labour.

31 **Labour variances: the two verification formulas?**

A LCV = LRV + idle time variance + LEV; LEV = labour mix variance + labour yield variance.

32 **Who is responsible for the labour rate variance and the labour efficiency variance (SM)?**

A Rate: the personnel manager, responsible for rate negotiation. Efficiency: the production manager or foreman.

33 **How is normal idle time of direct workers treated in standard costing?**

A An allowance for it is considered in setting the standard hours or the standard rate; abnormal idle time is shown separately.

34 **Formula: labour efficiency (efficiency ratio) as a percentage?**

A Standard hours for actual output ÷ actual hours worked × 100.

13.4 Variable overhead variances

35 **Formula: variable overhead cost variance?**

A Standard variable overheads for actual production − actual variable overheads, i.e. (SH × SR) − actual VOH.

36 **Formula: variable overhead expenditure (spending, or budget) variance?**

A (SR − AR) × AH, or (SR × AH) − (AR × AH): standard variable overheads for actual hours − actual variable overheads.

37 **Formula: variable overhead efficiency variance?**

A (SH − AH) × SR: standard variable overheads for production − standard variable overheads for actual hours.

38 **Verification of variable overhead variances?**

A VOH cost variance = VOH expenditure variance + VOH efficiency variance.

39 **Which actual hours do variable overhead variances use?**

A Actual hours worked (not hours paid); idle hours are excluded.

40 **Basis for variable overheads when the question says nothing (SM)?**

A Labour hours. If variable overheads consist of indirect materials, they vary with direct material used; machine hours if they depend on machine running.

41 **Standard variable overhead rate per hour, and standard hours for actual output?**

A Rate = budgeted variable overheads ÷ budgeted hours. SH = (budgeted hours ÷ budgeted output) × actual output.

42 **Only budgeted and actual output and overheads are given, no hours. Which VOH variance can be computed?**

A Only the cost variance: actual output × standard rate per unit − actual variable overheads.

43 **Labour efficiency variance ₹12,000 (A) at ₹60 an hour; VOH ₹15 an hour on the same hours. VOH efficiency variance?**

A ₹3,000 (A): 200 extra hours × ₹15.

13.5 Fixed overhead variances

44 **ICAI's definition: fixed overhead cost variance?**

A The difference between actual fixed overhead and absorbed fixed overhead: (SH × SR) – actual fixed overheads.

45 **Formula: fixed overhead expenditure variance?**

A Budgeted fixed overheads – actual fixed overheads, i.e. (BH × SR) – actual.

46 **Formula: fixed overhead volume variance?**

A Absorbed fixed overheads – budgeted fixed overheads, i.e. (SH × SR) – (BH × SR).

47 **Fixed overhead volume variance is divided into which three variances?**

A Efficiency variance, capacity variance and calendar variance.

48 **Formula: fixed overhead efficiency variance?**

A SR (SH − AH): absorbed fixed overhead − standard fixed overhead.

49 **Formula: fixed overhead capacity variance, and the revised one?**

A SR (AH − BH): standard − budgeted fixed overhead. With a calendar variance: SR (AH − revised budgeted hours).

50 **Formula: fixed overhead calendar variance?**

A Standard fixed overhead rate per day × (actual working days − budgeted working days).

51 **Revised budgeted hours?**

A Budgeted hours ÷ budgeted days × actual days.

52 **The fixed overhead "budget variance" is another name for which variance?**

A The fixed overhead expenditure variance: budgeted fixed overheads − actual fixed overheads (Fig 13.1: "Expenditure or Budget Variance").

53 **Overhead expenses variance (Illustration 12)?**

A Budgeted fixed overheads – (total actual overheads – standard variable overheads for actual output); with the production volume variance it makes up the total overhead cost variance.

54 **The two verification formulas for fixed overhead variances?**

A Cost variance = expenditure + volume; volume = efficiency + capacity + calendar.

55 **Fixed overheads ₹10,000; output 8,000 units against a budget of 10,000. Production volume variance?**

A (8,000 – 10,000) × ₹1 = ₹2,000 (A): the cost of failing to produce at the planned activity (SM 13.39).

13.6 Reconciling budgeted and actual profit, and accounting for variances

56 **What is the last step of the standard costing process (SM)?**

A Disposition of variances: transferring them to the relevant accounts (Costing Profit and Loss Account) as per the accounting method (plan) adopted.

57 **Controllable variances are best disposed of by transferring them to which account?**

A The Costing Profit and Loss Account.

58 **Treatment of a large under-absorption arising from a wrong estimate made for unavoidable reasons?**

A A supplementary rate, charged to cost of sales, finished goods and work-in-progress.

59 **Define controllable variance (SM).**

A A variance which can be controlled under normal operating conditions if a responsibility centre takes preventive measures and acts prudently.

60 **Define uncontrollable variance (SM).**

A A variance which occurs due to conditions beyond the control of a responsibility centre and cannot be controlled even though all preventive measures are in place.

61 **Define revision variance.**

A The difference between the original standard cost and the revised standard cost of actual production.

62 **The reconciliation of standard cost and actual cost?**

A Standard cost of actual output – net cost variance (F as plus, A as minus) = actual cost; net cost variance = material + labour + variable overhead + fixed overhead cost variances.

63 **Budgeted profit to flexible budget profit: what is the difference?**

A The effect of volume: (actual units – budgeted units) × standard contribution per unit (fixed overheads stay at budget).

64 **Income statement under standard costing: the layout?**

A Sales – standard cost of goods sold = standard gross margin; less adverse variances; add favourable variances; = actual gross margin.

65 **Material price variance on purchases, purchases above consumption: which actual figure does the reconciliation reach?**

A Actual purchases less the closing stock of material at standard price. On the consumption basis it reaches the actual cost of the material used.

66 **Why is closing stock carried at standard cost under standard costing?**

A Standard costs serve as the basis of inventory valuation; variances go to the period, so profit is not distorted by changes in stock values (SM 13.50).

CHAPTER 14

Marginal Costing

14.1 Marginal cost, contribution and the P/V ratio

1 **ICAI's definition of marginal cost (Chapter 1)?**

A The amount at any given volume of output by which aggregate costs increases if the volume of output is increased or decreased by one unit.

2 **ICAI's definition of marginal costing?**

A A costing system where products or services and inventories are valued at variable costs only; it does not take consideration of fixed costs (also called direct costing).

3 **Contribution: ICAI's definition?**

A The difference between sales revenue and total variable costs, irrespective of manufacturing or non-manufacturing.

4 **The marginal cost equation?**

A S − V = C = F ± P (sales − variable cost = contribution = fixed cost ± profit or loss).

5 **P/V ratio: the two formulas?**

A Contribution ÷ sales × 100; or change in contribution (or profit) ÷ change in sales × 100.

6 **Under marginal costing, which costs are product (inventoriable) costs?**

A Variable manufacturing costs only: direct materials, direct employee cost, direct expenses and variable manufacturing overheads.

7 **Are variable selling and distribution costs included in the value of closing stock under marginal costing?**

A No. They are deducted in arriving at contribution but excluded from the valuation of finished goods and WIP.

8 **In the "profit-volume ratio", what does the word "profit" actually refer to?**

A It is a misnomer: it refers to contribution (sales revenue – variable costs), not to profit after fixed costs.

9 **CVP analysis provides information about which five matters (ICAI)?**

A Behaviour of cost in relation to volume; the volume at which the business breaks even; sensitivity of profits to variation in output; profit for a projected sales volume; production and sales for a target profit.

10 **The seven advantages of marginal costing, in ICAI's order?**

A Simplified pricing policy; proper recovery of overheads; shows realistic profit; how much to produce; more control over expenditure; helps in decision making; short-term profit planning.

11 **The seven limitations of marginal costing, in ICAI's order?**

A Difficulty in classifying fixed and variable elements; dependence on key factors; scope for low profitability; faulty valuation; unpredictable nature of cost; ignores time factor and investment; understating of WIP.

14.2 Break-even point, margin

of safety and angle of incidence

12 **Formula: break-even point in units?**

A Fixed cost ÷ contribution per unit (contribution per unit = selling price − variable cost per unit).

13 **Formula: break-even sales in rupees, two ways?**

A Fixed cost ÷ P/V ratio, or (fixed cost ÷ total contribution) × sales.

14 **What is the cash break-even point, and what is left out of fixed cost for it?**

A The break-even point calculated only with fixed costs payable in cash: cash fixed costs ÷ contribution per unit. Depreciation and other non-cash fixed costs are excluded.

15 **ICAI's definition of margin of safety?**

A The margin between sales and the break-even sales; it can be indicated in quantitative or monetary terms.

16 **Margin of safety from profit and P/V ratio?**

A Margin of safety = profit ÷ P/V ratio (so profit = margin of safety × P/V ratio).

17 **Formula: margin of safety ratio?**

A (Total sales − break-even sales) ÷ total sales × 100.

18 **ICAI's definition of angle of incidence?**

A The angle formed by the intersection of the sales line and the total cost line at the break-even point; it shows the rate at which profit is earned once the break-even point is reached.

19 **What does a large angle of incidence with a high margin of safety indicate?**

A An extremely favourable position.

20 **P/V ratio from two periods' sales and profit?**

A Change in profit ÷ change in sales × 100, valid when price, unit variable cost and fixed cost are unchanged; a loss counts as a negative profit.

21 **Sales needed for a desired profit, in units and in rupees?**

A Units = (fixed cost + desired profit) ÷ contribution per unit; sales value = (fixed cost + desired profit) ÷ P/V ratio.

22 **Effect of an increase in fixed cost on the P/V ratio, the BEP and the angle of incidence?**

A P/V ratio: no change. BEP: rises. Angle of incidence: no change.

14.3 Break-even and profit-volume charts

23 **What does a break-even chart record on each axis (ICAI)?**

A Costs and revenues on the vertical axis and the level of activity on the horizontal axis.

24 **The three lines of a conventional break-even chart and where each starts?**

A Fixed cost (horizontal at F), total cost (from F on the vertical axis), sales (from the origin).

25 **How does a contribution break-even chart differ from a conventional one?**

A It shows a variable cost line (from the origin) instead of the fixed cost line, so contribution can be read as the gap between the sales and variable cost lines.

26 **ICAI's stated major limitation of the conventional break-even chart?**

A It cannot be used to measure contribution.

27 **On a profit-volume chart, what is the loss at nil activity?**

A Equal to the fixed cost.

28 **On a profit-volume chart, where is the break-even point?**

A Where the profit line intersects the horizontal (zero profit) axis.

29 **The biggest advantage of the profit-volume chart (ICAI)?**

A It depicts clearly the effect on profit and the break-even point of any changes in the variables.

30 **Slope of the profit line on a P/V chart?**

A Contribution per unit with units on the horizontal axis; the P/V ratio with sales value on it.

31 **Effect of an increase in fixed cost on the profit-volume line?**

A It moves down parallel to itself; the break-even point moves to the right.

32 **Volume at which two options earn equal profit?**

A Difference in fixed costs ÷ difference in contribution per unit.

33 **In a multi-product profit graph, which line gives the break-even point?**

A The straight profit line from the loss at zero (fixed cost) to the final total profit, whose slope is the overall P/V ratio; not the profit path.

14.4 Marginal versus absorption costing and profit reconciliation

34 **ICAI's definition of absorption costing?**

A The practice of charging all costs, both variable and fixed, to operations, processes or products.

35 **Product (inventoriable) costs under marginal and under absorption costing?**

A Marginal costing: variable manufacturing costs. Absorption costing: total manufacturing costs (variable and fixed).

36 **Under absorption costing, when are selling and administration overheads charged?**

A As expenses when incurred (period costs), like under marginal costing.

37 **Profit when closing stock is more than opening stock?**

A Absorption costing profit is higher, because part of the fixed overhead in closing stock is carried forward to the next period.

38 **Profit when opening stock is more than closing stock?**

A Marginal costing profit is higher, because fixed cost of the preceding period enters this period's cost of goods sold through opening stock.

39 **Profit when opening and closing stocks are equal?**

A Equal under both methods, provided the fixed cost element in both stocks is the same amount.

40 **Formula: difference between absorption and marginal profit?**

A Fixed overhead in closing stock – fixed overhead in opening stock.

41 **Under- or over-absorption of fixed production overhead: on which units, and in which statement?**

A Actual fixed production overhead − units produced × absorption rate; adjusted to cost of goods sold in the absorption costing statement only.

42 **In ICAI's absorption costing income statement, at what value is opening stock and closing stock taken?**

A Opening stock at the cost of the previous period's production; closing stock at the production cost of the current period.

43 **The main difference between marginal and absorption costing is in the treatment of which cost?**

A Fixed (production) overheads.

44 **The five points of distinction between marginal and absorption costing (headings)?**

A Inventory valuation; fixed cost as period cost vs charged to production; contribution highlighted vs conventional net profit; effect of stock levels on unit cost; unit cost constant vs falling with production.

14.5 Limiting factor and product mix

45 **ICAI's definition of limiting factor?**

A Limiting factor is anything which limits the activity of an entity. The factor is a key to determine the level of sale and production, thus it is also known as key factor.

46 **Supply-side limiting factors listed by the Study Material?**

A Men (employees), materials (raw material or supplies), machine (capacity) and money (availability of fund or budget).

47 **Demand-side limiting factors listed by the Study Material?**

A Demand for the product, and other factors like nature of product, regulatory and environmental requirements.

48 **ICAI formula for profitability with a key factor?**

A Profitability = Contribution ÷ Key factor, i.e. contribution per unit of the limiting factor.

49 **With a scarce resource, how are products ranked?**

A By contribution per unit of the limiting factor (per hour, per kg), highest first; not by contribution or profit per unit of product.

50 **Ranking measure when total sales value is the key factor?**

A P/V ratio (contribution per rupee of sales).

51 **How are machine hours per unit found when only variable overhead per unit and the rate per machine hour are given?**

A Hours per unit = variable overhead per unit ÷ variable overhead rate per machine hour.

52 **In what order is a limited resource allocated?**

A Minimum commitments first, then products in rank order up to each one's maximum demand; the last product gets the balance.

53 **Make or buy with scarce machine hours: ranking measure?**

A Saving per machine hour = (buy price − variable cost of making) ÷ machine hours per unit; items cheaper to buy are bought outright.

54 **Which limitation of marginal costing concerns key factors?**

A Dependence on key factors: contribution of a product itself is not a guide for optimum profitability unless it is linked with the key factor.

14.6 Make or buy, shut down and other short-term decisions

55 **When is a cost relevant for decision making, according to the Study Material?**

A Only if (a) it is a future cost and (b) it differs under the two options under consideration.

56 **ICAI's four steps of the framework for decision making?**

A Identification of problem; identification of options; evaluation of the options; selection of the option.

57 **The two principles for identifying costs and benefits, and the two for estimating them?**

A Identification: controllability and relevance. Estimation: variability and traceability.

58 **Historical, sunk, committed, opportunity and notional costs: relevant or not?**

A Historical, sunk and committed: irrelevant. Opportunity: relevant. Notional (imputed): relevant only if the company is actually forgoing a benefit.

59 **ICAI's definition of shut down costs (Chapter 1)?**

A Costs which continue to be incurred even when a plant is temporarily shut down, e.g. rent, rates, depreciation; all fixed costs which cannot be avoided during the temporary closure of a plant.

60 **Formula: shut-down point in units?**

A (Total fixed cost − shut-down cost) ÷ contribution per unit, where shut-down cost = unavoidable fixed costs + additional costs of closing.

61 **Make or buy: what is the relevant cost of making?**

A Variable cost of making + fixed costs avoidable if bought + contribution forgone on any alternative use of the capacity.

62 **When should a product be discontinued on cost grounds?**

A When the fixed costs avoided by dropping it (plus any income from the freed capacity) exceed the contribution it earns; apportioned common fixed costs are ignored.

63 **Special order without enough idle capacity: what is added to its variable and specific costs?**

A The contribution lost on the regular sales it displaces (an opportunity cost).

64 **ICAI's definition of opportunity cost?**

A The value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action.

65 **Formula: cost indifference point between two alternatives?**

A Differential fixed cost ÷ differential variable cost per unit.

14.7 Multi-product break-even analysis

66 **What does the Study Material say about break-even in a multi-product environment?**

A The break-even formula needs adjustments: the contribution is calculated by taking weights for the products; the weights may be of sales mix quantity or sales mix values.

67 **Formula: composite contribution per unit when the mix is given in units?**

A Σ (contribution per unit × units of the product in the mix ÷ total units in the mix).

68 **Formula: composite break-even point?**

A Common fixed cost ÷ composite contribution per unit; each product's BEP = composite BEP × its share of units.

69 **Formula: composite P/V ratio when the mix is given in values?**

A Σ (P/V ratio × sales mix %), ICAI's "contribution per rupee of sales"; equals total contribution ÷ total sales.

70 **Formula: composite break-even sales?**

A Fixed cost ÷ composite P/V ratio; each product's BES = composite BES × its sales mix %.

71 **Which CVP assumption does multi-product break-even rest on?**

A The analysis either covers a single product or assumes that the proportion of the products sold remains constant as total units sold change.

72 **Effect on break-even sales of a mix shift towards high-P/V products (sales and fixed costs unchanged)?**

A Composite P/V rises, so break-even sales fall and profit rises.

73 **ICAI's second method in Illustration 11 for each product's BEP?**

A Share the fixed cost in the ratio of weighted contributions, then each product's BEP = its share of fixed cost ÷ its contribution per unit.

74 **What is the composite break-even batch?**

A Fixed cost ÷ contribution of one batch of the sales mix (e.g. 5 ball pens + 3 gel pens); multiply by each product's units per batch for its break-even units.

CHAPTER 15

Budgets and Budgetary Control

15.1 Budgets, budgetary control, the budget manual and key factor

1 **CIMA's definition of a budget?**

A A financial and/or quantitative statement, prepared and approved prior to a defined period of time, of the policy to be pursued during that period for the purpose of attaining a given objective. It may include income, expenditure and employment of capital.

2 **CIMA's definition of budgetary control?**

A The establishment of budgets relating to the responsibilities of executives to the requirements of a policy, and the continuous comparison of actual with budgeted results, either to secure by individual action the objective of that policy or to provide a basis for its revision.

3 **Budgetary control involves which four things?**

A Establishment of budgets; continuous comparison of actuals with budgets; revision of budgets when circumstances change; fixation of responsibility for failure to achieve the budget targets.

4 **What is a key factor in budgeting?**

A A factor which sets a limit to the total activity (usually one or two, sometimes more), such as non-availability of power or lack of demand; it must be identified and its influence estimated while preparing the budget.

5 **CIMA's definition of a budget manual?**

A A document which sets out the responsibilities of the persons engaged in, the routine of, and the forms and records required for, budgetary control.

6 **The six steps in the preparation of budgets, in order?**

A Defining objectives; identifying the key budget factor; appointing the controller or officer; the budget manual; the budget period; the standard of activity or output.

7 **The five steps for establishing budgetary control?**

A Determine objectives and policies; determine the activities; draw up a plan for each class of activity in quantitative and monetary terms; lay out a system of comparison of actual with budget and find causes of variation; ensure corrective action or revise the plan.

8 **The four components of a budgetary control system?**

A Physical budgets, cost budgets, profit budgets and financial budgets (for example, cash budget, capital expenditure budget, budgeted balance sheet).

9 **Who decides the budget period, and what is it normally?**

A The Budget Committee; normally a calendar year or a period co-terminus with the financial year, divided into months, quarters or trading periods. There is no general rule.

10 **ICAI's six limitations of budgetary control?**

A Based on estimates; time factor; co-operation required; expensive; not a substitute for management; rigid document.

11 **Forecast versus budget in one line?**

A A forecast is an assessment of probable future events and denotes flexibility; a budget is a commitment, a definite target which management seeks to attain on the basis of the forecasts.

12 **Long-term, short-term and current budgets: their periods?**

A Long-term: more than a year, usually three to ten years; short-term: generally one or two years; current: months and weeks, related to current conditions.

15.2 Functional budgets and the master budget

13 **CIMA's definition of the master budget?**

A The summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed.

14 **Format: production budget in units?**

A Budgeted sales + desired closing stock = total quantity required; less opening stock = units to be produced.

15 **Format: direct material purchase budget?**

A Units required for production + desired closing stock = total requirement; less opening stock = units to be purchased; × unit price = purchase cost.

16 **The production budget is based on which four things?**

A The sales budget; the production capacity of the factory; the planned increase or decrease in finished stocks; the policy governing outside purchase.

17 **Stabilised production versus stabilised stocks?**

A Stabilised production: capacity fully utilised, inventory carrying cost varies with stocks held. Stabilised stocks: carrying cost lowest, but capacity may be under-utilised.

18 **The four purposes of the plant utilisation budget?**

A Determine the load on each process or machine group; show overloaded ones for corrective action; dovetail sales and production budgets where capacity cannot be raised; boost sales where capacity is surplus.

19 **Two considerations that govern the material purchase budget?**

A The economic order quantity, and the re-order point with safety stocks to cover fluctuations in demand.

20 **The three merits of the direct labour budget?**

A It defines the direct and indirect labour force required; lets the personnel department plan recruitment and training so labour turnover is minimised; reveals the labour cost for the manufacturing cost and cash budgets.

21 **Hours to be paid when the efficiency ratio is 80% and down-time is 20% of productive hours?**

A Productive hours = standard hours ÷ 0.80; hours paid = productive hours × 1.20. Hours beyond the normal hours are overtime.

22 **Cost of goods sold budget versus production cost budget?**

A Both add direct materials, direct labour and factory overhead; the cost of goods sold budget also adds opening and deducts closing finished goods. Without those adjustments it is the production cost budget.

23 **Four methods of allocating research and development expenses?**

A A percentage of total sales value; a percentage of net profit; a total sum from past experience and future R&D plans; a sum fixed by the cash resources available.

24 **Is a purchases budget the same as a materials budget?**

A No: a purchases budget is a wider concept and may include purchases of items other than materials.

15.3 Flexible budgets

25 **CIMA's definition of a fixed budget?**

A A budget designed to remain unchanged irrespective of the level of activity actually attained.

26 **CIMA's definition of a flexible budget?**

A A budget which, by recognizing the difference between fixed, semi-variable and variable costs, is designed to change in relation to the level of activity attained.

27 **Flexible budget allowance at the actual level?**

A Fixed cost + variable cost per unit × actual activity (semi-variable costs split into their fixed and variable parts first).

28 **High and low points method: the two formulas?**

A Variable cost per unit = change in expense ÷ change in output; fixed cost = total expense − variable cost per unit × output.

29 **In which three situations may flexible budgeting be resorted to?**

A A new business venture whose demand is hard to forecast; a business dependent on the fluctuations of nature; a labour-intensive industry whose production depends on the availability of labour.

30 **Five cases ICAI lists as suitable for a flexible budget?**

A Seasonal fluctuations in sales or production; frequent new products or design changes; make-to-order industries like ship building; industries influenced by fashion; general changes in sales.

31 **Three demerits of a flexible budget?**

A It needs a proper accounting system and perfect knowledge of the factors of production and business circumstances; it requires a standard costing system; it is very expensive and labour oriented.

32 **Merits and demerits of a fixed budget?**

A Merits: very simple to understand, less time consuming. Demerits: misleading in a dynamic organisation; unsuitable for a long period; unsuitable when conditions change constantly; accurate estimates not possible.

33 **Fixed vs flexible budget: the five points of difference?**

A Change (rigid vs re-cast); levels (one vs several); variance analysis (not useful vs useful); cost ascertainment and pricing (misleading vs facilitated); comparison of actual with budget (meaningless when levels differ vs meaningful).

34 **Why is applying the budgeted cost per unit to a different activity level wrong?**

A Because fixed costs do not change with activity: the cost per unit falls as activity rises, so the allowance must be fixed cost + variable rate × actual activity.

35 **How are buses, trucks or batches treated in a flexible budget?**

A As step (semi-variable) costs: the number needed is rounded up at each level and costed whole.

15.4 Cash budget

36 **ICAI's definition of a cash budget?**

A A detailed budget of cash receipts and cash payments incorporating both revenue and capital items for the budget period, analysed to show the monthly flow of cash so that the balance can be forecasted at regular intervals.

37 **The two parts of a cash budget, with examples?**

A Cash receipts (cash sales, collections from debtors, sale of assets, borrowings) and cash disbursements (cash purchases, payments to creditors, employees' remuneration, bonus, advances to suppliers, capital expenditure).

38 **Under which component of a budgetary control system does the cash budget fall?**

A Financial budgets, with the capital expenditure budget and the budgeted balance sheet.

39 **Which budgets and policies affect the cash budget?**

A The operating budgets (sales, direct materials purchases, selling and administrative expenses) and the capital expenditure budget, dividend policies and plans for equity or long-term debt financing.

40 **ICAI's four objectives of preparing a cash budget?**

A Assess the probable cash position (excess or shortage, so borrowing or investment is arranged in advance); coordinate cash with total working capital, sales, investment and debt; establish a sound basis for credit and current control of cash; reveal the effect of seasonal requirements, large stocks and delayed collections on the cash position.

41 **ICAI's four advantages of a cash budget?**

A Helps secure optimum working capital and plan payments to shareholders; eases the strains of a cash shortage; facilitates temporary investment of excess cash; provides for normal growth.

42 **Four items that never enter a cash budget?**

A Depreciation (and amortisation), bad debts written off, provisions, and the profit or loss on sale of an asset (only its proceeds).

43 **Receipts from debtors when only balances are given?**

A Opening debtors + credit sales − closing debtors (− bad debts written off).

44 **The three methods of preparing a cash budget (FM Study Material)?**

A Receipts and payments method; adjusted income method; adjusted balance sheet method.

45 **How much is borrowed when the balance before financing is below the minimum cash balance?**

A The minimum balance minus the balance before financing (from minus ₹90,000 to a minimum of ₹2,00,000 is ₹2,90,000), so the minimum is restored, not merely zero.

15.5 Zero based budgeting and performance budgeting

46 **ICAI's definition of zero-based budgeting?**

A A method of budgeting which requires each cost element to be specifically justified, though the activities to which the budget relates are not being undertaken for the first time; without justification, the budget allowance is zero.

47 **The four stages of ZBB, in order?**

A Identification and description of decision packages; evaluation of decision packages; ranking (prioritisation) of decision packages; allocation of resources.

48 **What are decision packages in ZBB?**

A The programmes or activities for which a decision is required, described for technical specifications, financial impact (cost-benefit analysis) and other issues such as environmental, regulatory and social.

49 **On what factors are ZBB decision packages evaluated?**

A Synchronisation with organisational objectives, availability of funds, regulatory requirement, etc.

50 **Why is ZBB also called priority-based budgeting?**

A Because the evaluated decision packages are ranked by the priority of the activities before resources are allocated.

51 **For which costs is ZBB best suited in a company?**

A Discretionary costs such as research and development, training programmes and advertisement.

52 **ICAI's two limitations of ZBB?**

A Creating and ranking decision packages on new data is very tedious for management; the activities selected are based on traditional functional departments, so the scheme may not be implemented properly.

53 **Traditional budgeting versus ZBB: who justifies the spending?**

A In traditional budgeting top management decides why an amount is spent on a decision unit; in ZBB the responsibility shifts to the manager of the decision unit.

54 **What does a performance budget present?**

A The purposes and objectives for which funds are required, the costs of the programmes proposed to achieve them, and quantitative data measuring the accomplishments and work performed under each programme.

55 **Performance budgeting is considered synonymous with what, and why?**

A Responsibility accounting: it fixes the responsibility of each executive in terms of results and appraises performance continuously.

56 **The ARC's three basic steps in performance budgeting?**

A A meaningful functional, programme and activity classification of government operations; accounting and financial management brought in line with it; norms, yardsticks, work units and unit costs evolved under each programme and activity for reporting and evaluation.

57 **Five things an enterprise must ensure to adopt performance budgeting?**

A Objectives spelt out in concrete terms; translated into functions, programmes, activities and tasks within fiscal constraints; realistic norms and performance indicators in quantifiable physical units; a decentralised responsibility structure; an accounting and reporting system to monitor and review actual performance against budgets.

15.6 Control ratios, budget variances, motivation, feedback and feed-forward

58 **Formula: efficiency ratio?**

A Standard hours for actual output ÷ actual hours worked × 100.

59 **Formula: activity ratio?**

A Standard hours for actual output ÷ budgeted hours × 100.

60 **Formula: calendar ratio?**

A Available working days ÷ budgeted working days × 100.

61 **Standard capacity usage ratio against actual capacity usage ratio?**

A Standard: budgeted hours ÷ maximum possible hours in the budget period × 100. Actual: actual hours worked ÷ maximum possible working hours × 100.

62 **Formula: actual usage of budgeted capacity ratio (the capacity ratio)?**

A Actual working hours ÷ budgeted hours × 100.

63 **How are the efficiency, activity and capacity ratios related?**

A Activity ratio = efficiency ratio × capacity ratio; so efficiency = activity ÷ capacity (capacity ratio = actual ÷ budgeted hours).

64 **When is a budget ratio favourable, per ICAI?**

A When it is 100% or more; below 100% the performance is considered unfavourable.

65 **A kulhad café's repairs cost is ₹25,000 under budget for the quarter because the planned servicing of its chai machines was put off. Does the favourable variance show efficient performance?**

A No. A favourable variance does not necessarily indicate efficient performance: this one arises from not carrying out an expense provided in the budget, so it is a postponement, not a saving (SM 15.72–15.73).

66 **The cold-chain fleet's fuel cost is 15% below budget, a significant deviation. Should it appear in a budget report prepared on the principle of exception?**

A Yes. Such a report shows both favourable and unfavourable variances; "exception" does not mean adverse only (SM 15.73).

67 **Define controllable and uncontrollable variances.**

A Controllable: can be controlled under normal operating conditions if the responsibility centre takes preventive measures and acts prudently. Uncontrollable: arise from conditions beyond its control and cannot be controlled even with all preventive measures in place.

68 **Feedback against feed-forward control, one line each?**

A Feedback: actuals compared with budget after the completion of the budget period, then corrective action, responsibility fixed and future targets modified (ex-post corrective; less expensive). Feed-forward: actuals continuously monitored during the period and targets reviewed and reset if necessary (ex-ante preventive; needs a robust MIS with integrated ERP; very expensive; suits dynamic environments).

69 **ICAI's four considerations that make budgeting motivating?**

A Performance measurement; achievable targets; optimum utilisation of resources; involvement in the budgeting process.